

BFS Traversal

Step 1: Start at A

Legend

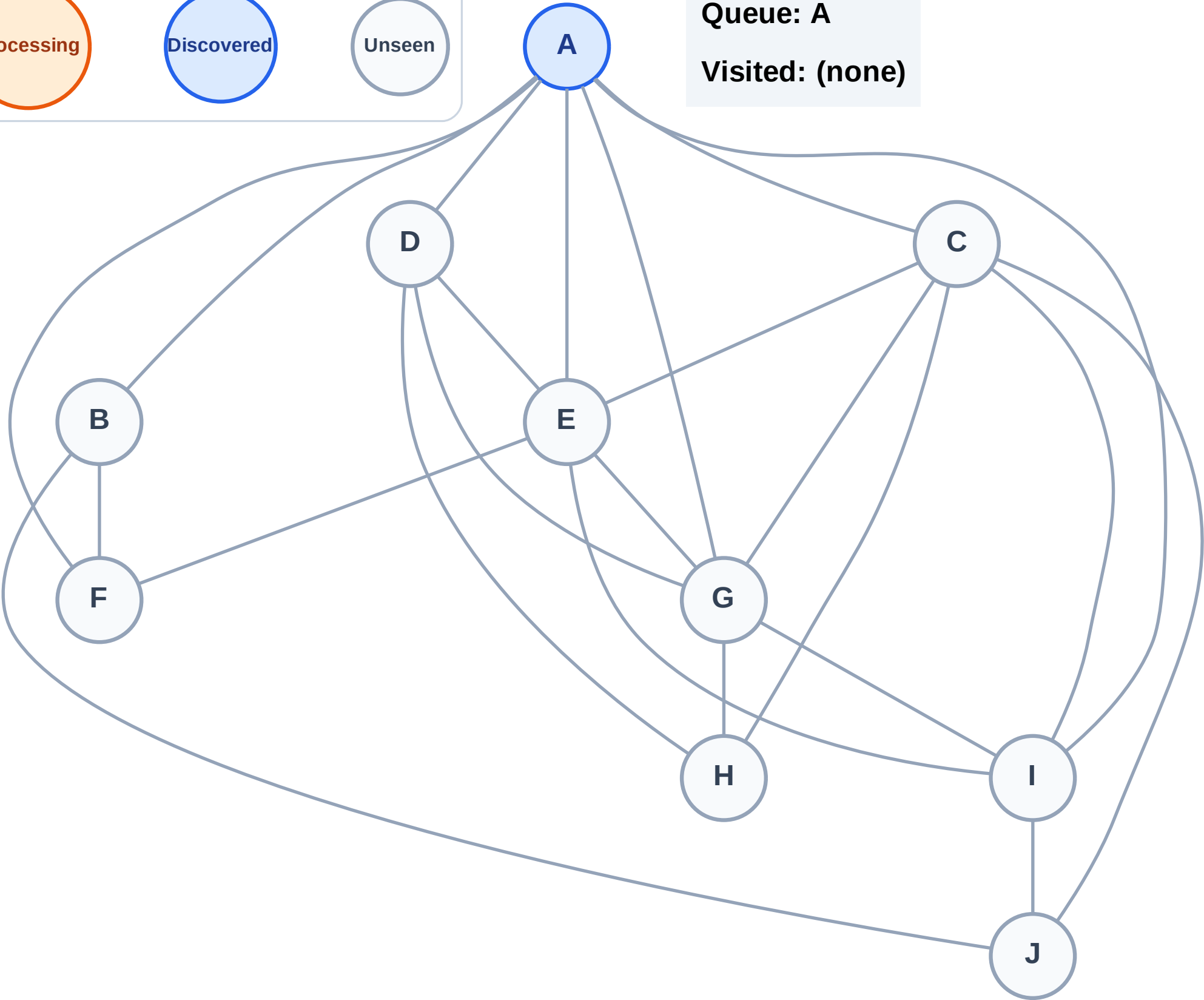
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

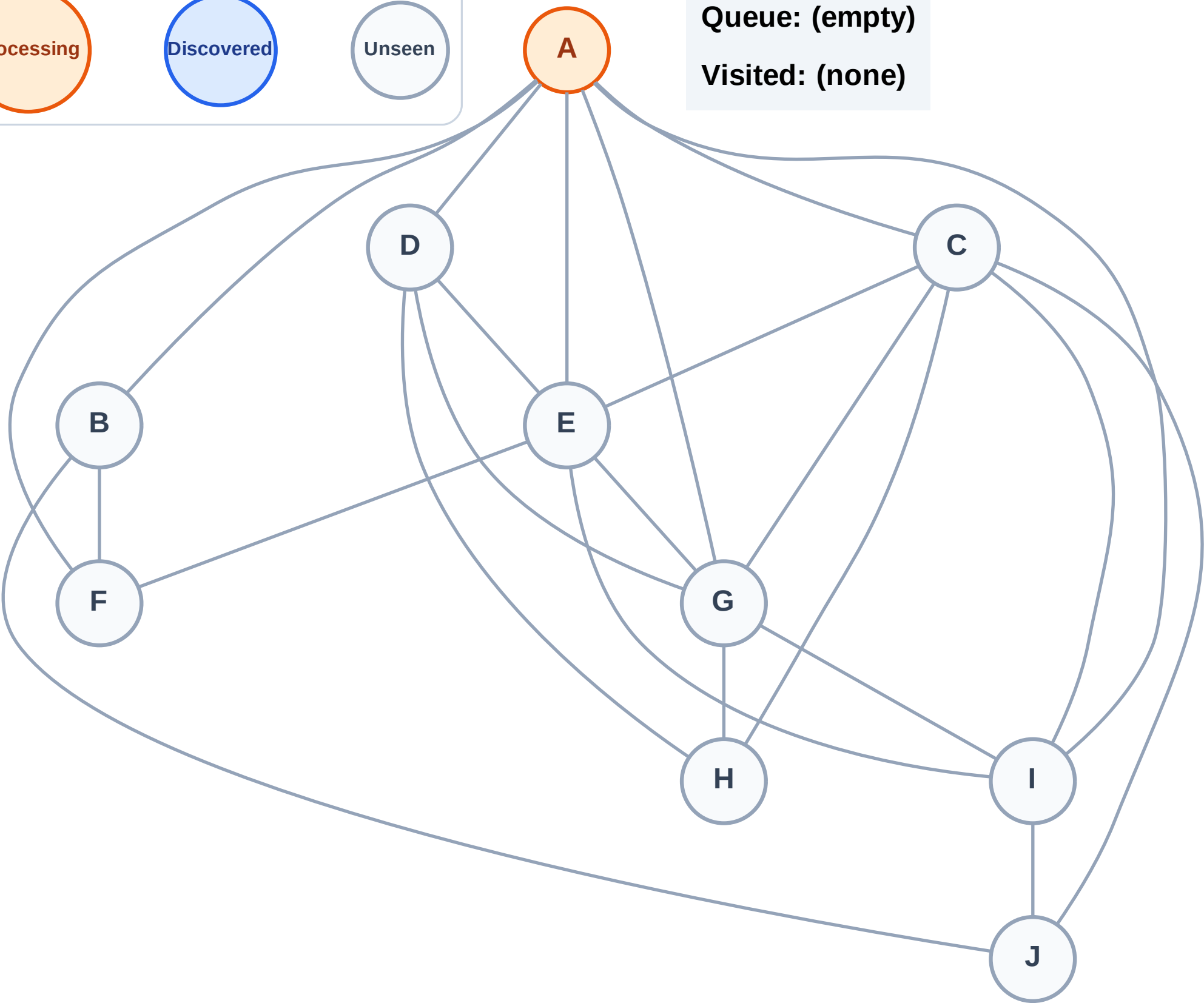
Step 2: Process node A

Legend



Queue: (empty)

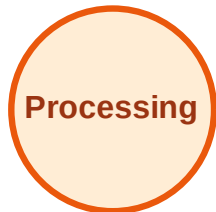
Visited: (none)



BFS Traversal

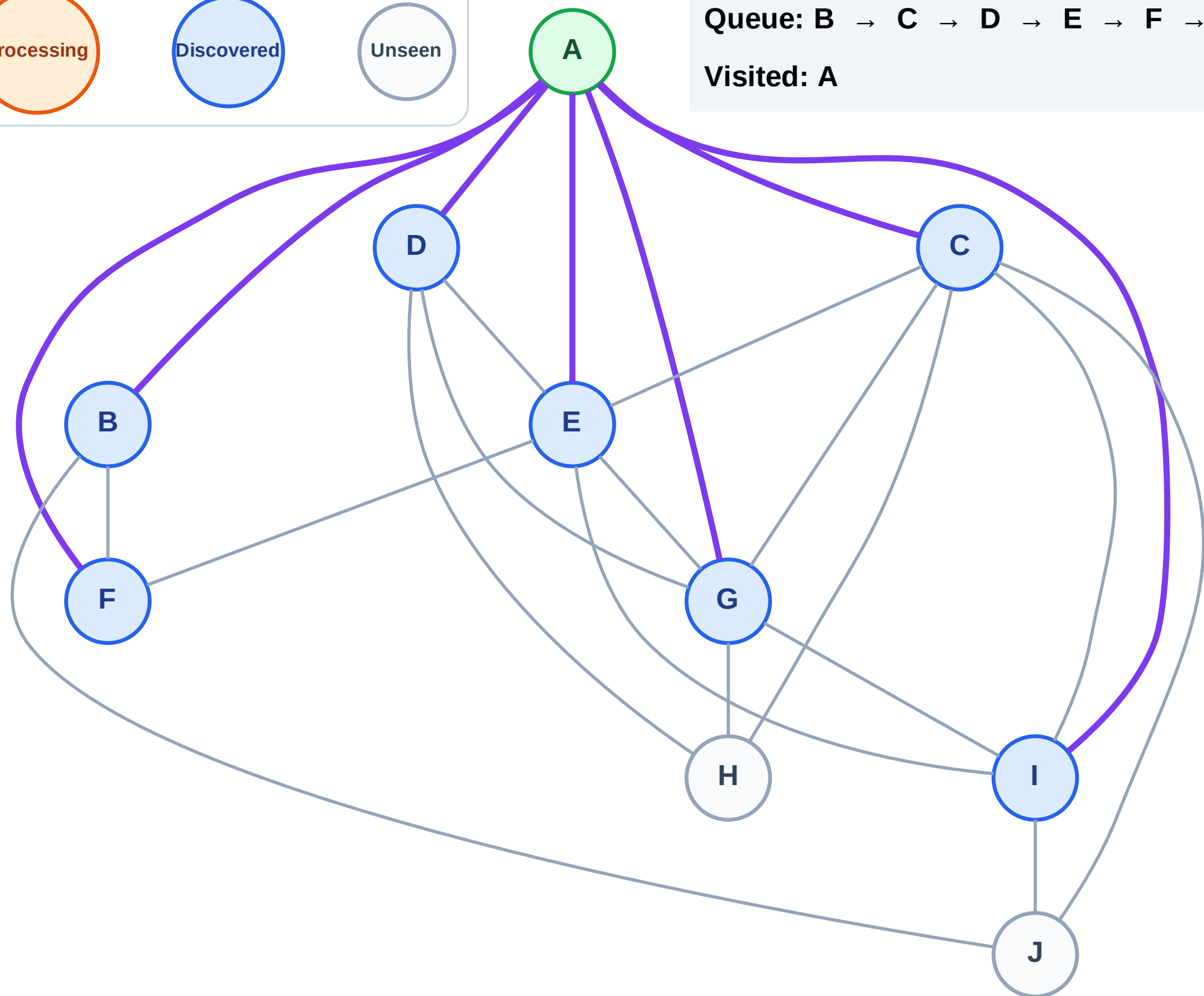
Step 3: Discover B, C, D, E, F, G, I from A

Legend



Queue: B → C → D → E → F → G → I

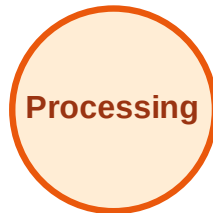
Visited: A



BFS Traversal

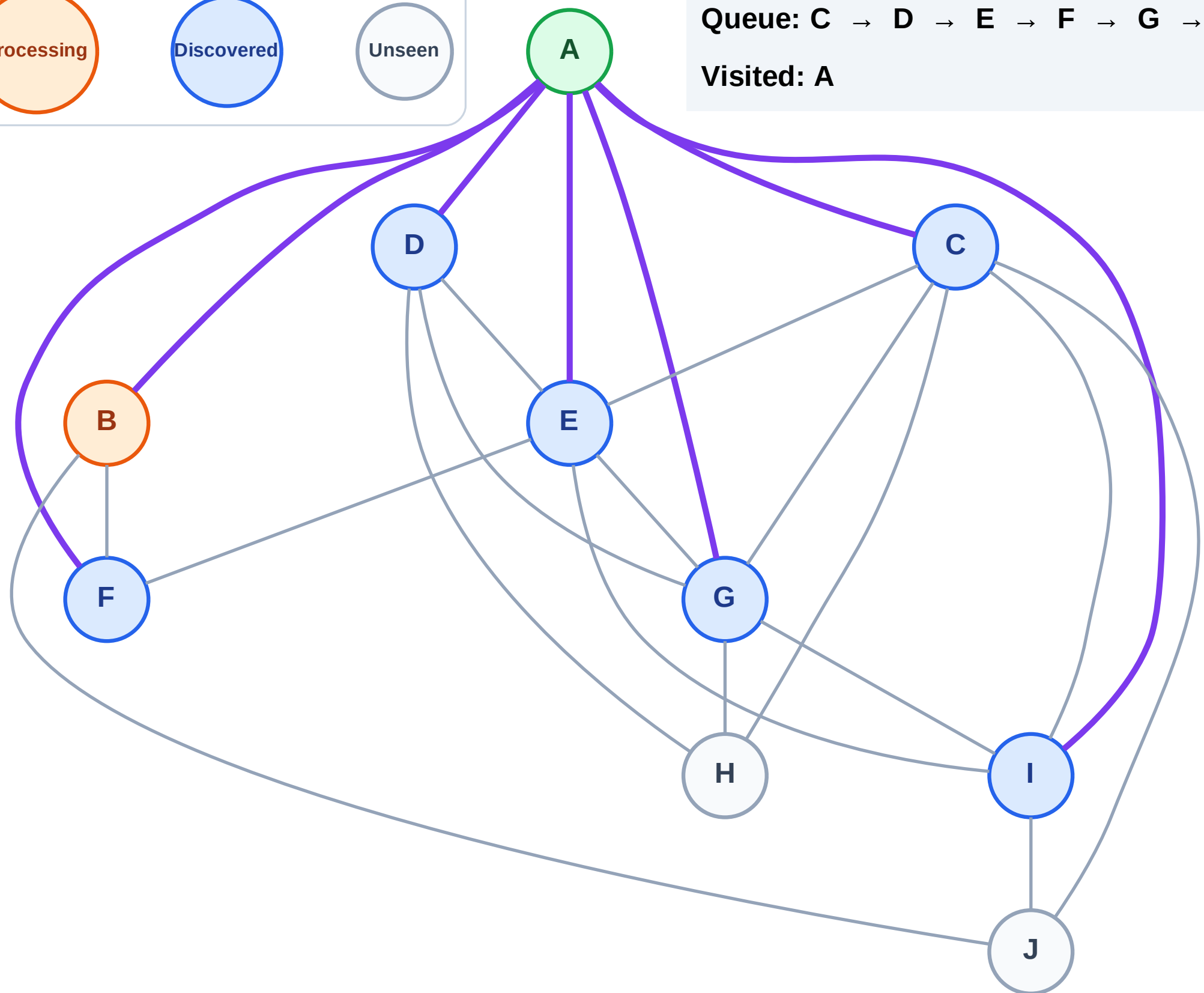
Step 4: Process node B

Legend



Queue: C → D → E → F → G → I

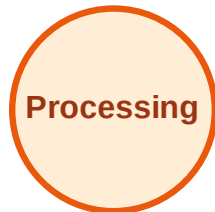
Visited: A



BFS Traversal

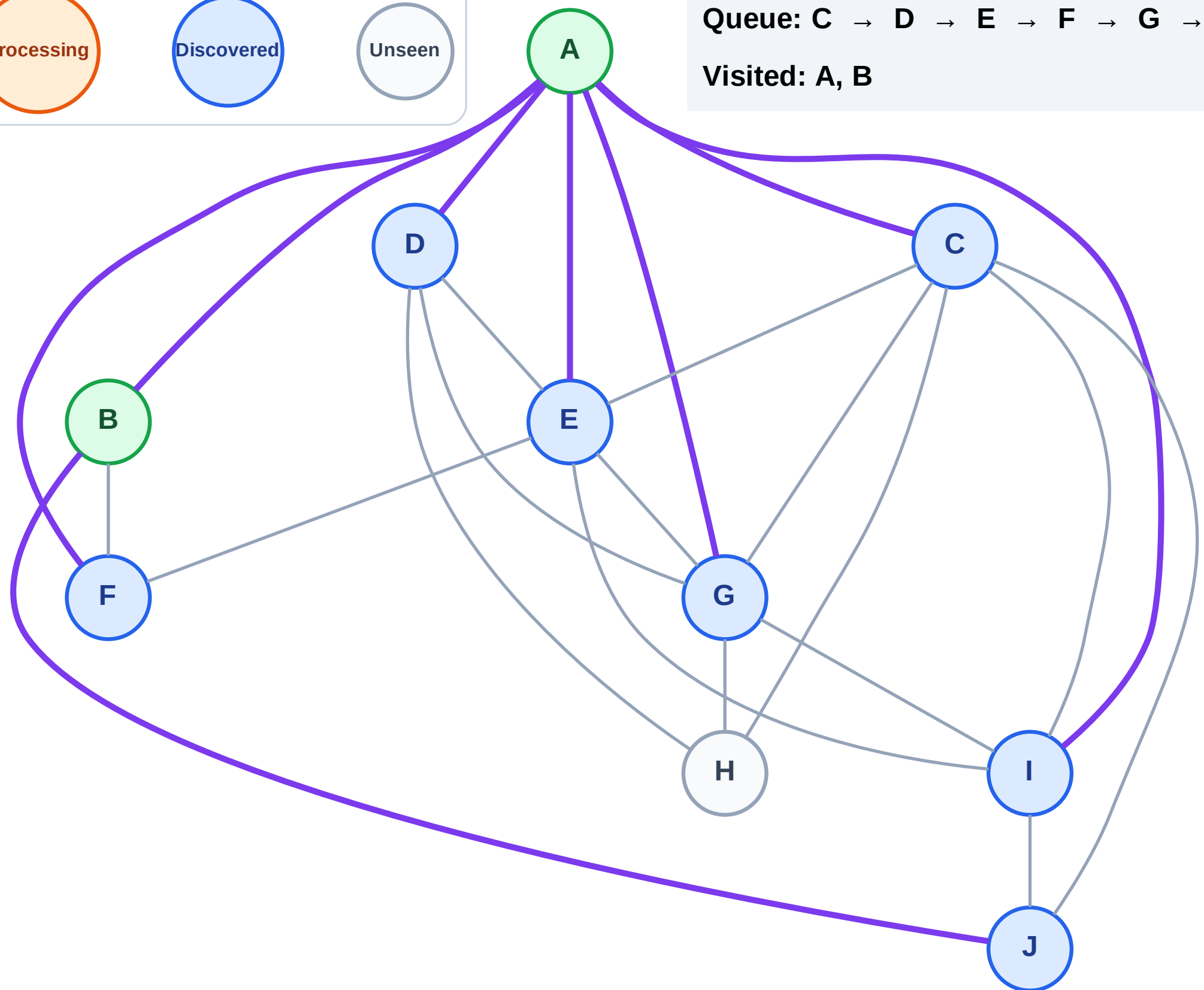
Step 5: Discover J from B

Legend



Queue: C → D → E → F → G → I → J

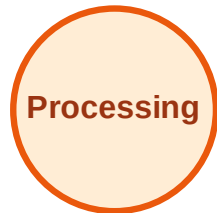
Visited: A, B



BFS Traversal

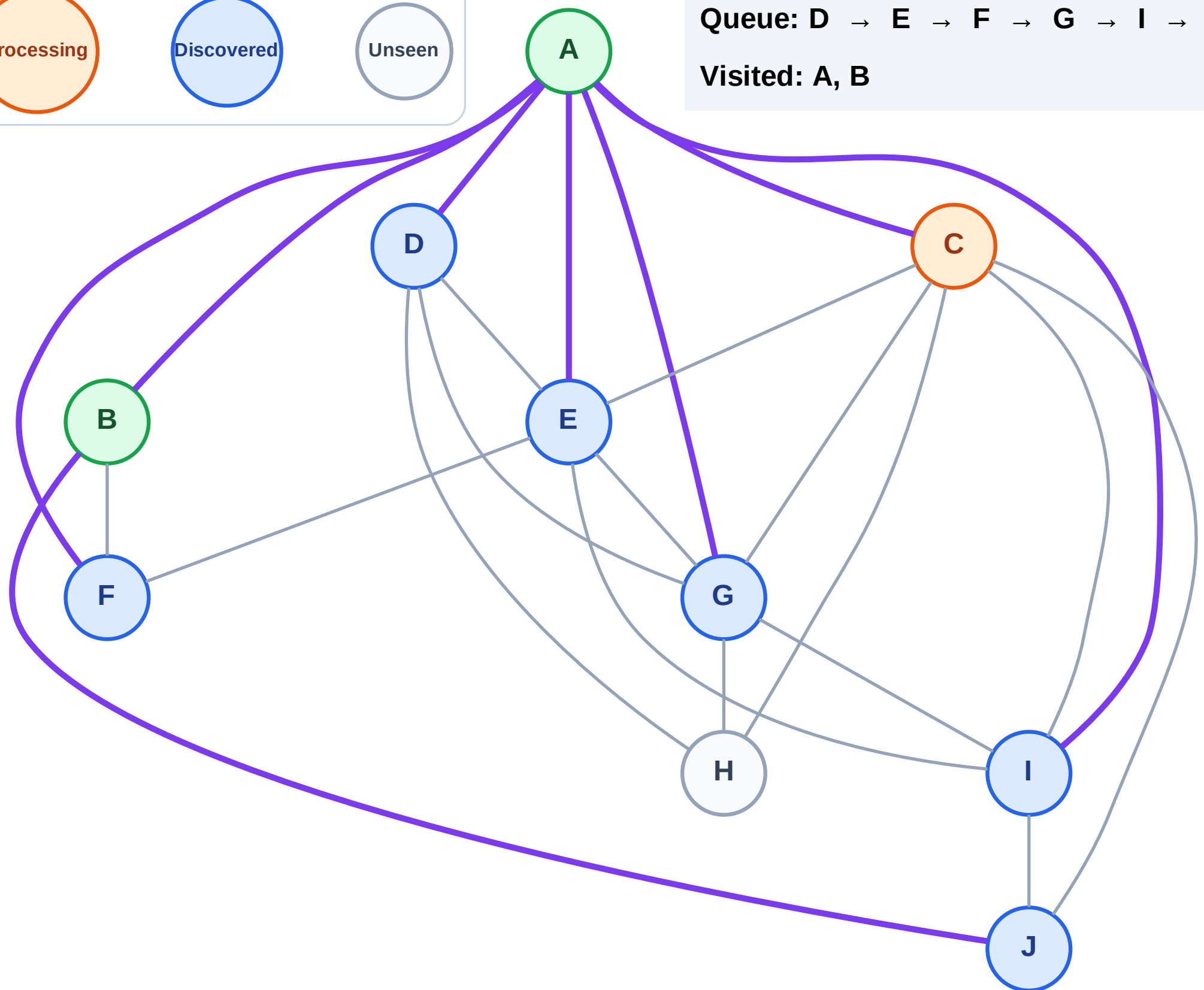
Step 6: Process node C

Legend



Queue: D → E → F → G → I → J

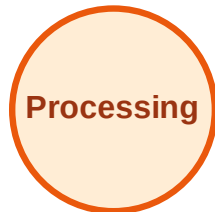
Visited: A, B



BFS Traversal

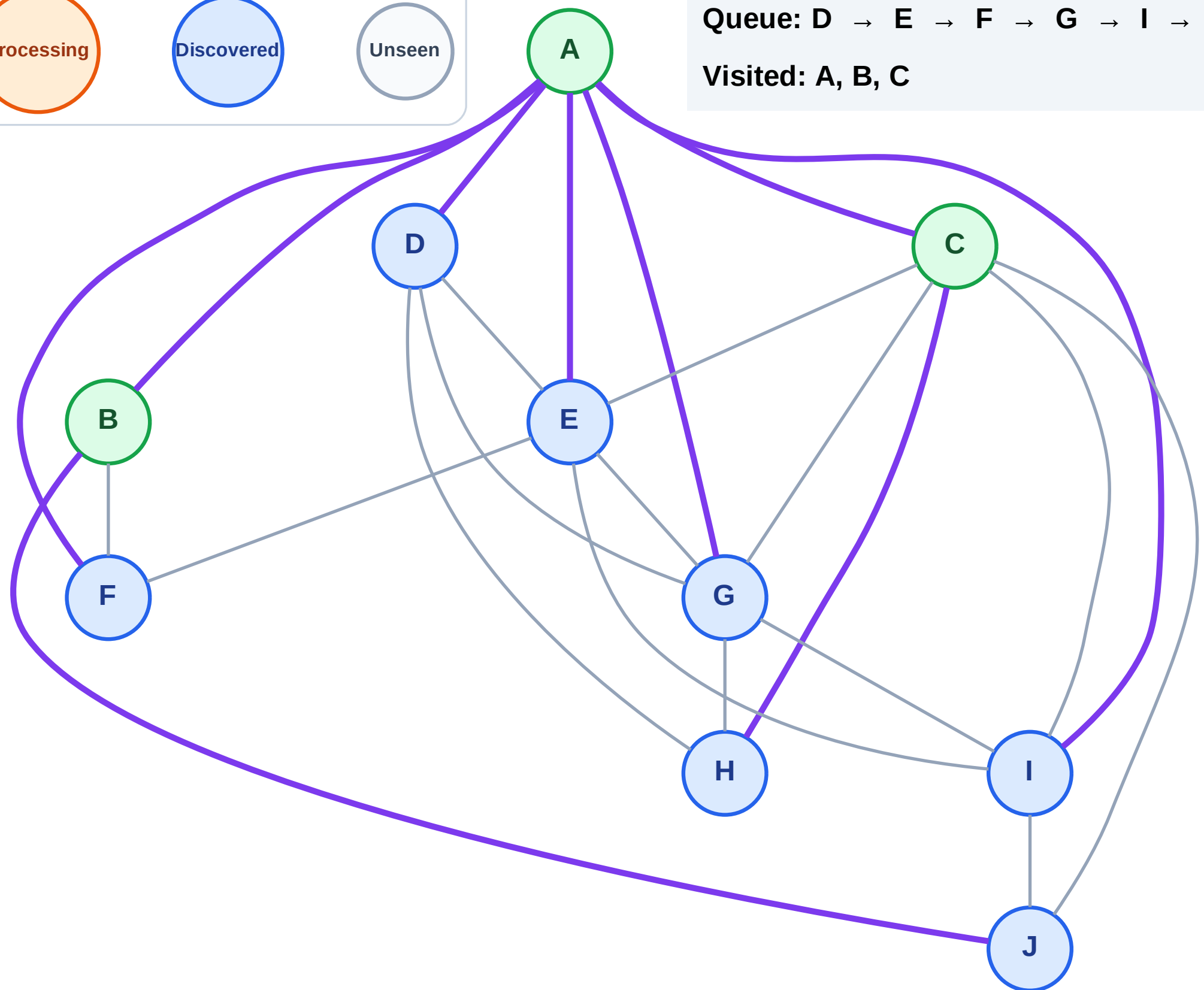
Step 7: Discover H from C

Legend



Queue: D → E → F → G → I → J → H

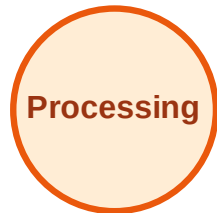
Visited: A, B, C



BFS Traversal

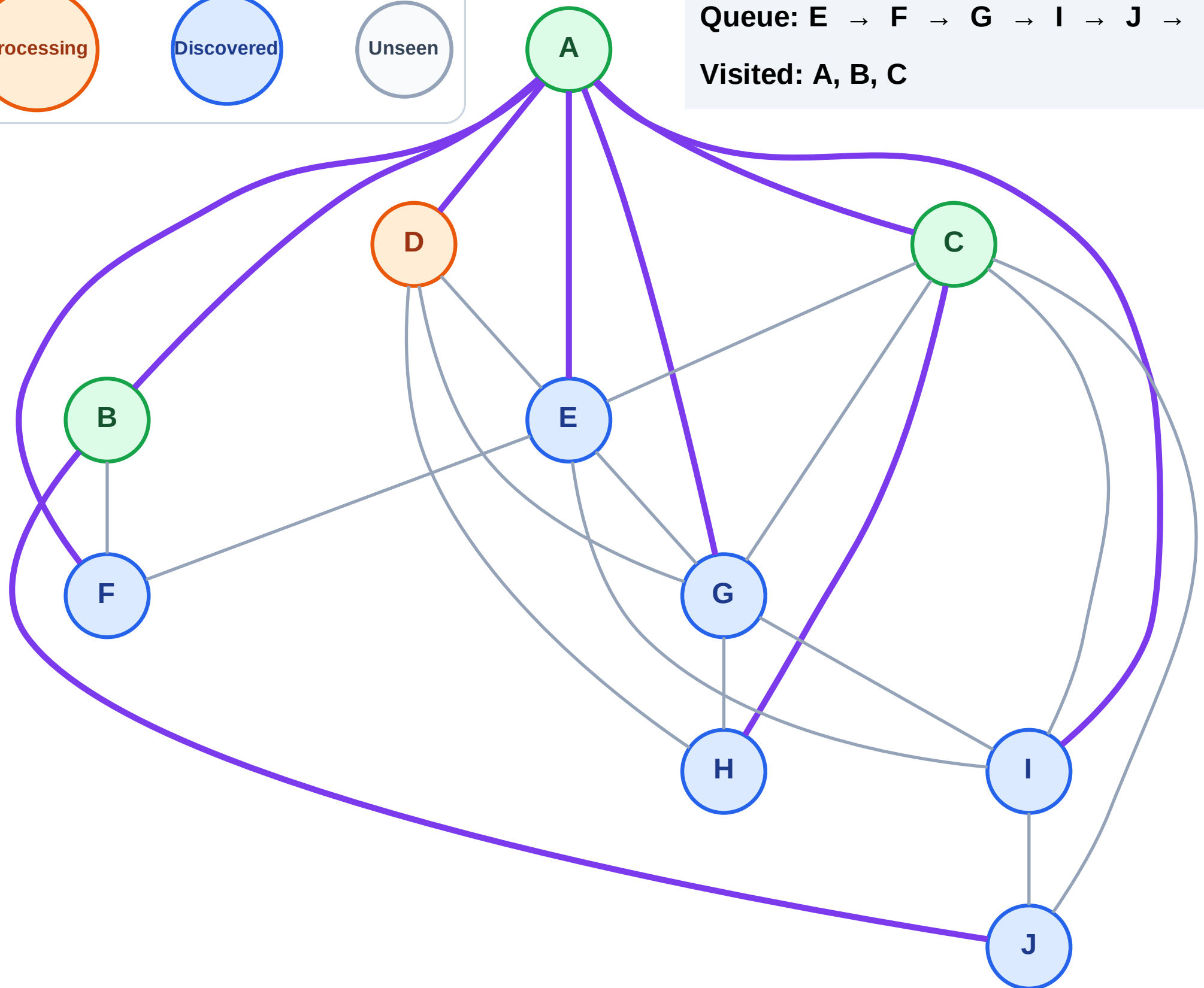
Step 8: Process node D

Legend



Queue: E → F → G → I → J → H

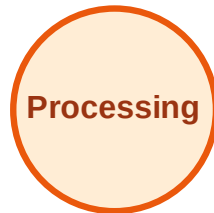
Visited: A, B, C



BFS Traversal

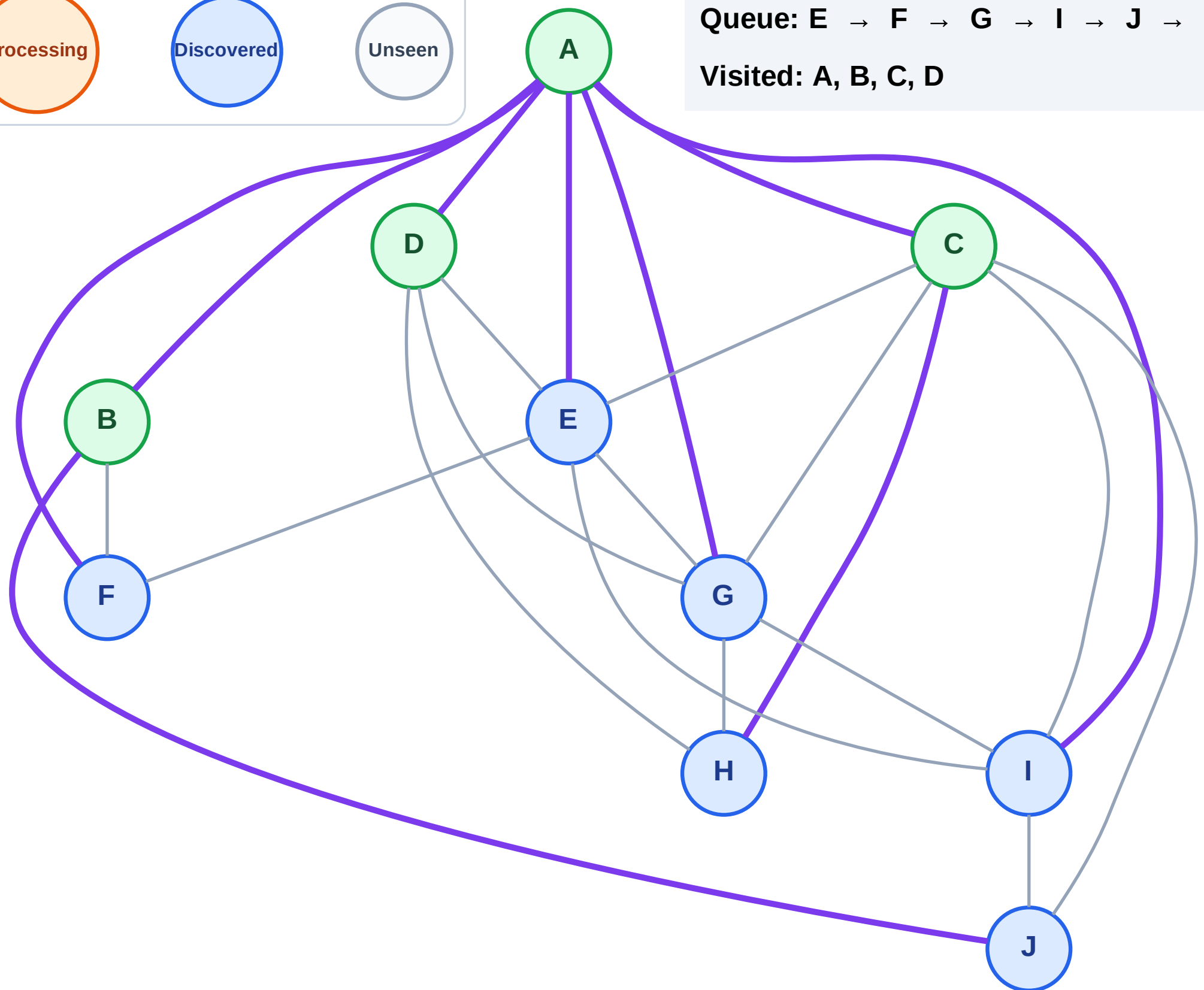
Step 9: No new neighbors from D

Legend



Queue: E → F → G → I → J → H

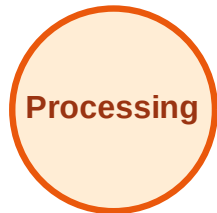
Visited: A, B, C, D



BFS Traversal

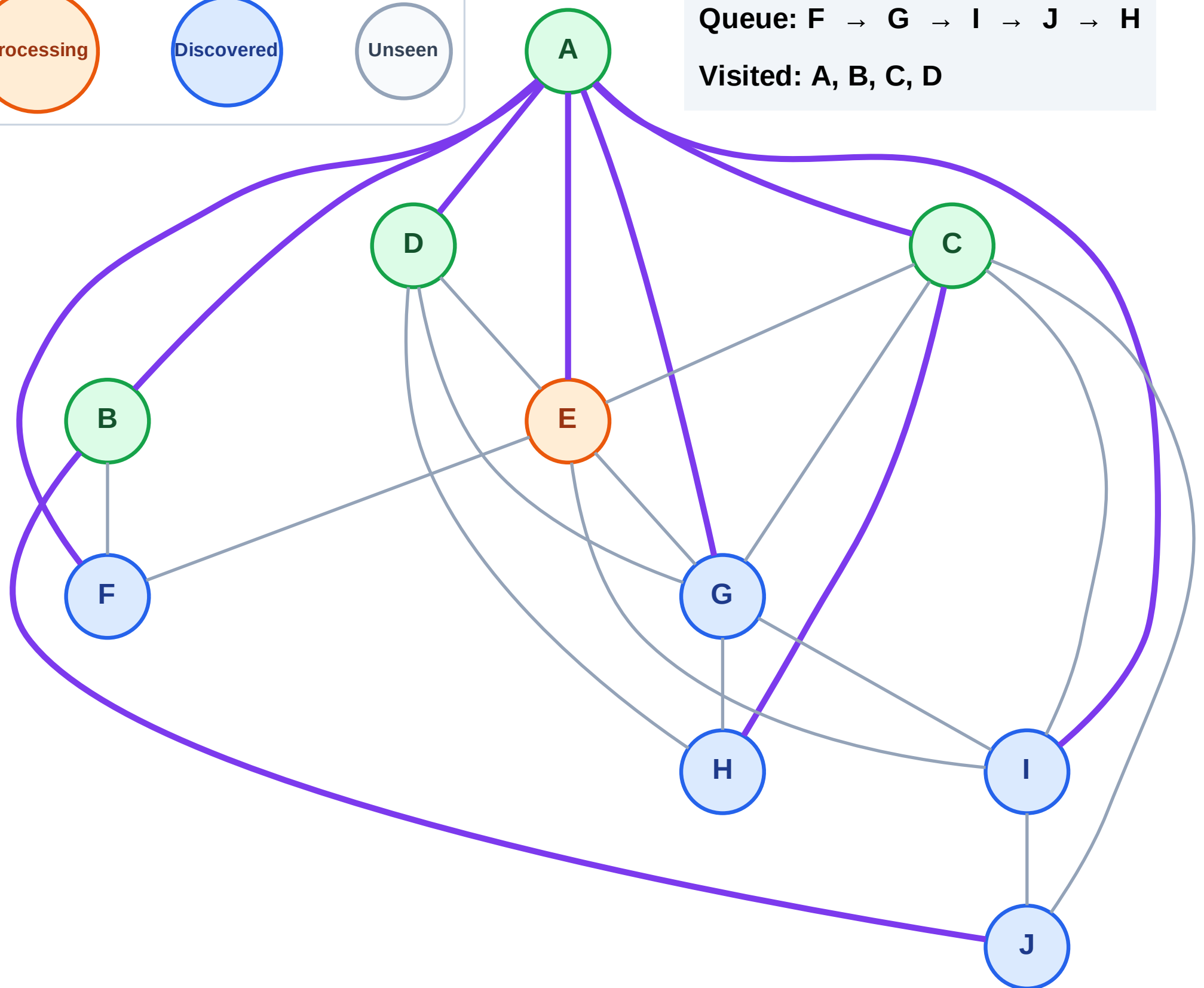
Step 10: Process node E

Legend



Queue: F → G → I → J → H

Visited: A, B, C, D



BFS Traversal

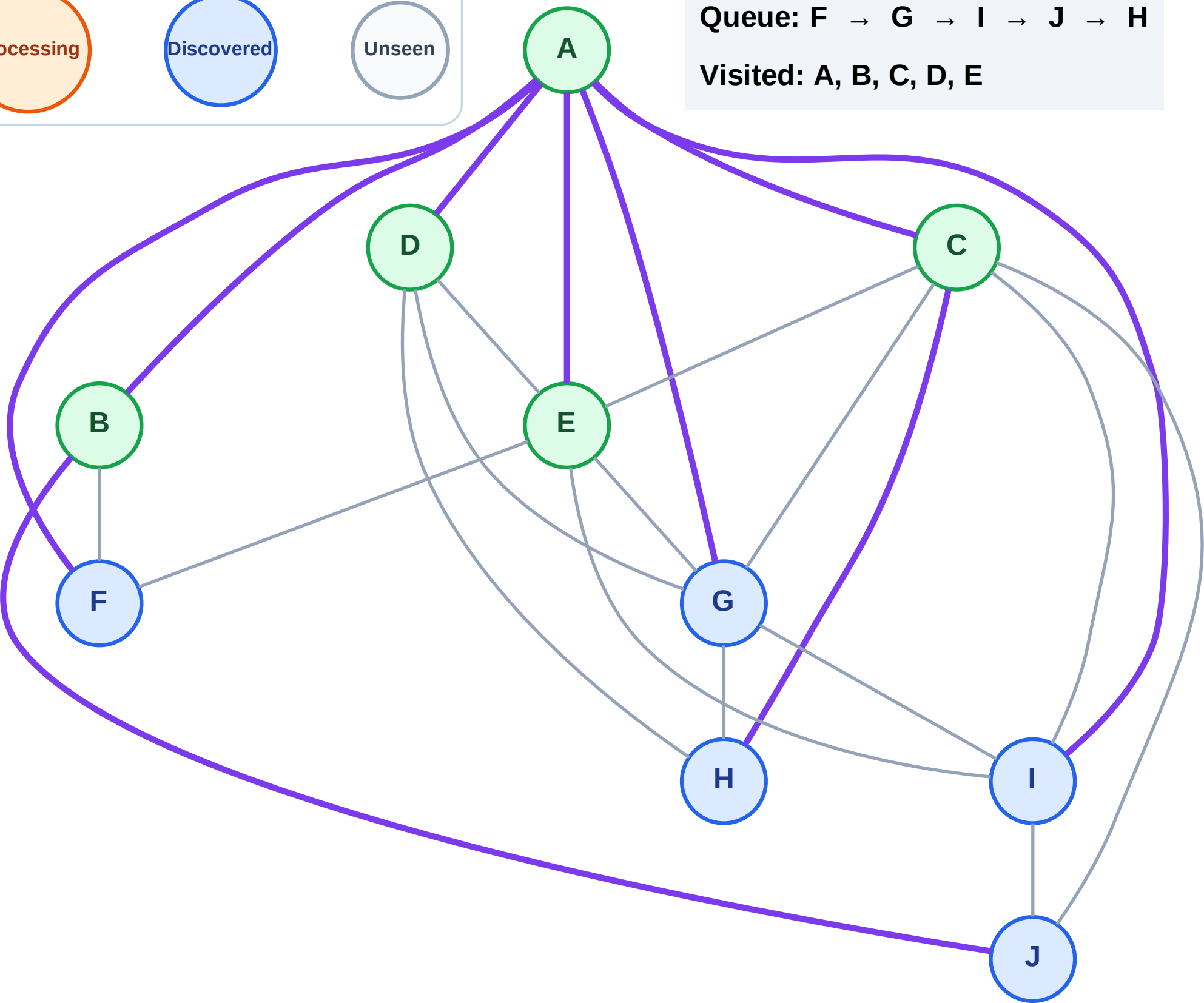
Step 11: No new neighbors from E

Legend



Queue: F → G → I → J → H

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node F

Legend

Complete

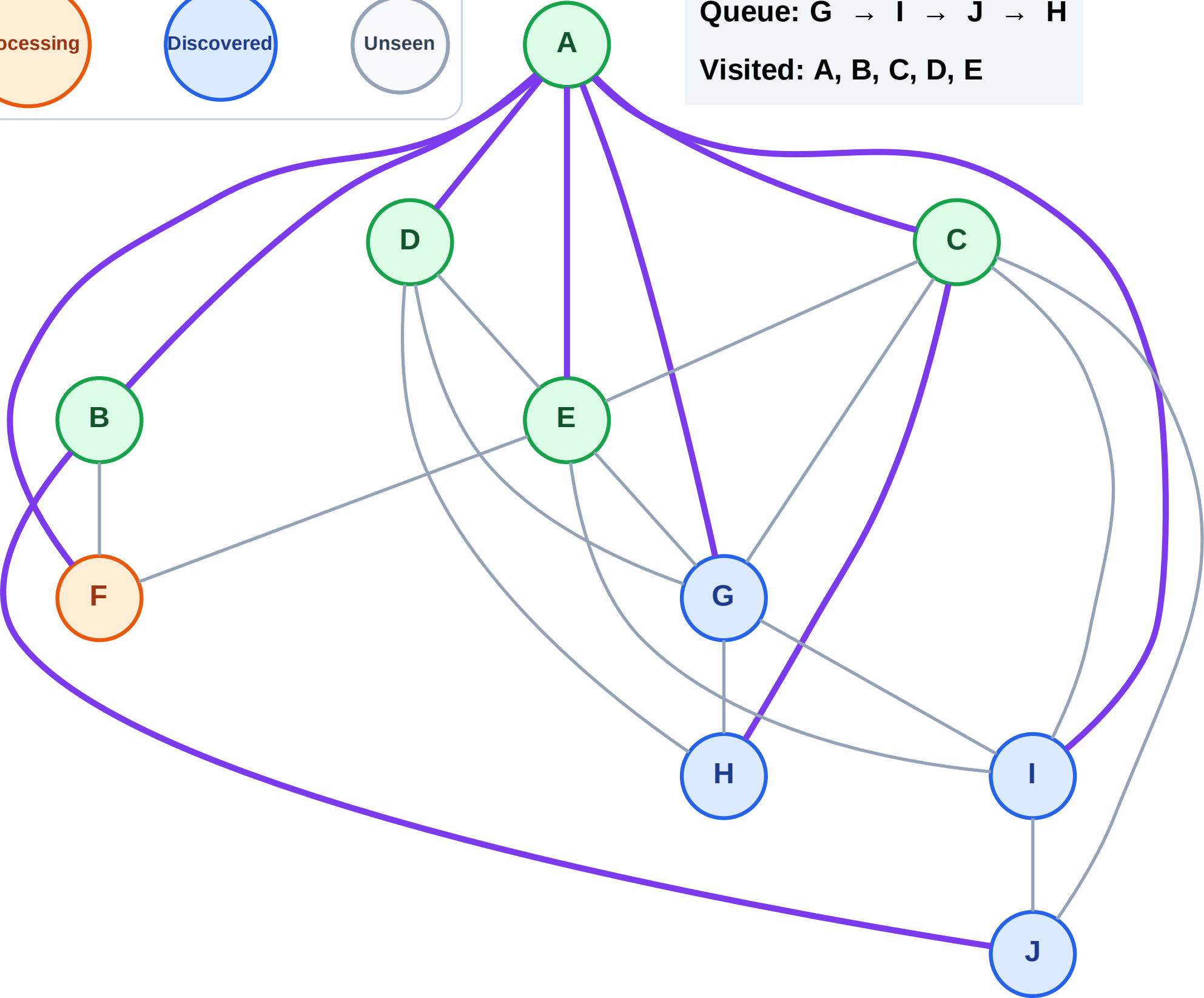
Processing

Discovered

Unseen

Queue: G → I → J → H

Visited: A, B, C, D, E



BFS Traversal

Step 13: No new neighbors from F

Legend

Complete

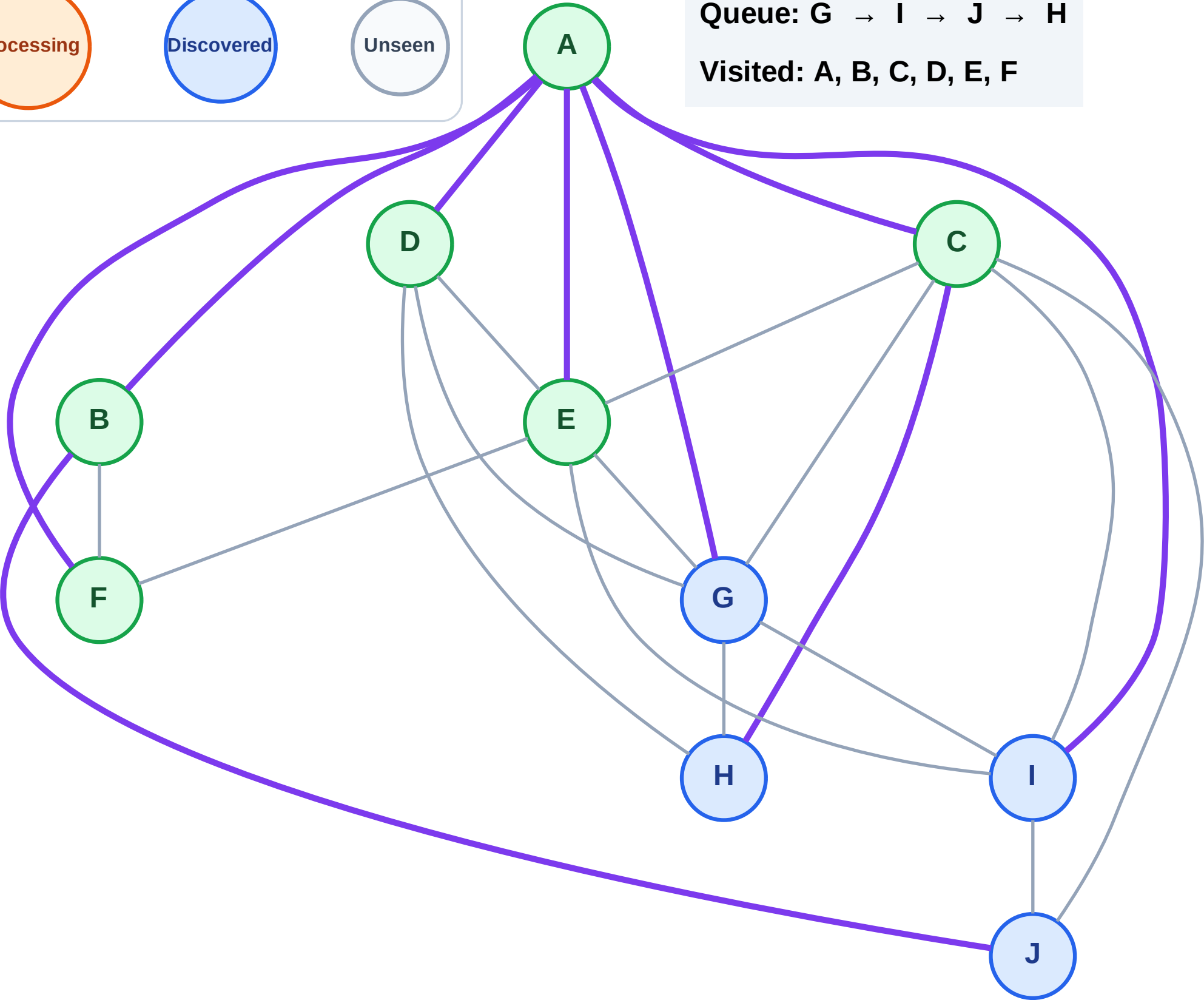
Processing

Discovered

Unseen

Queue: G → I → J → H

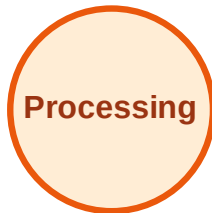
Visited: A, B, C, D, E, F



BFS Traversal

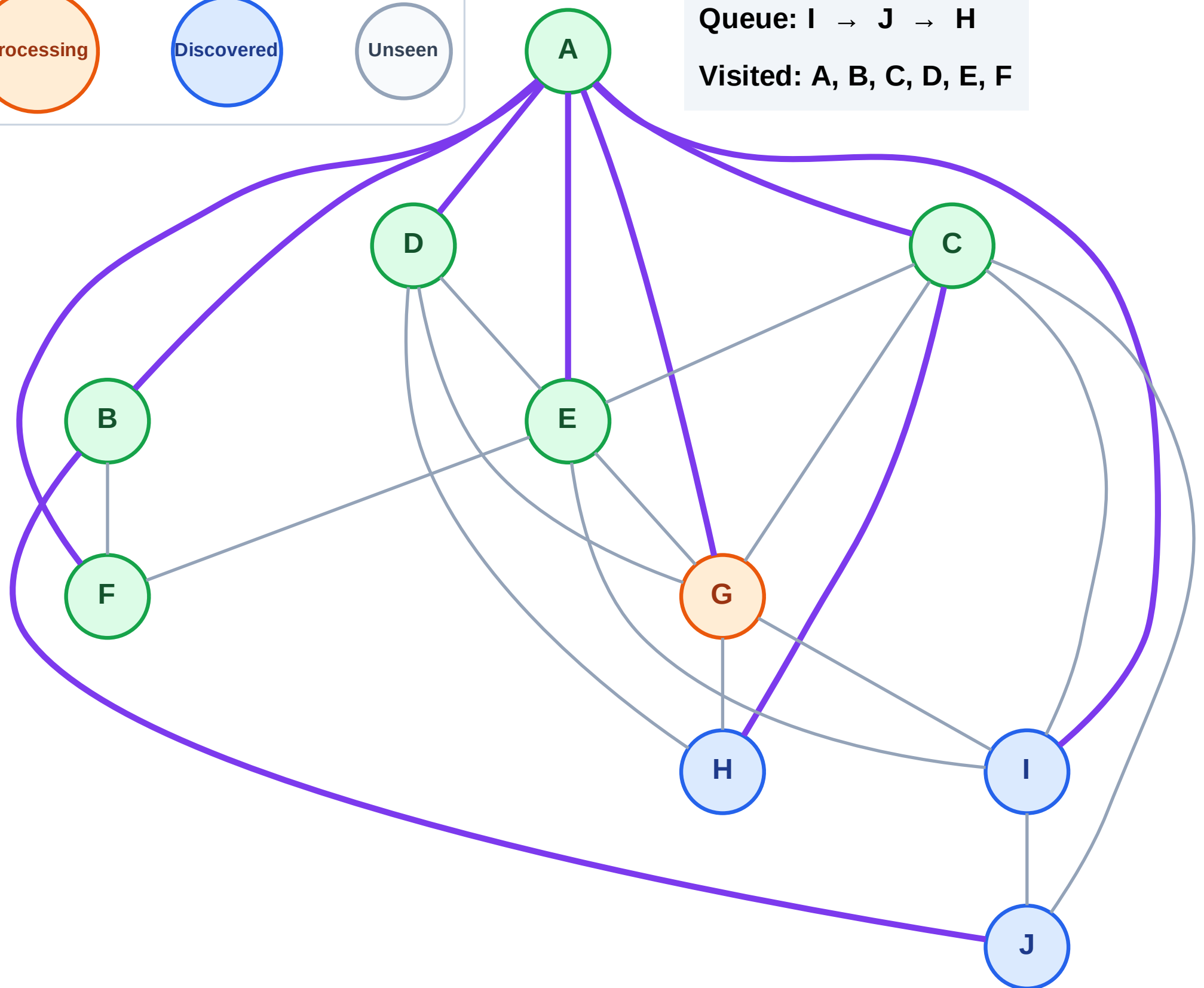
Step 14: Process node G

Legend



Queue: I → J → H

Visited: A, B, C, D, E, F



BFS Traversal

Step 15: No new neighbors from G

Legend

Complete

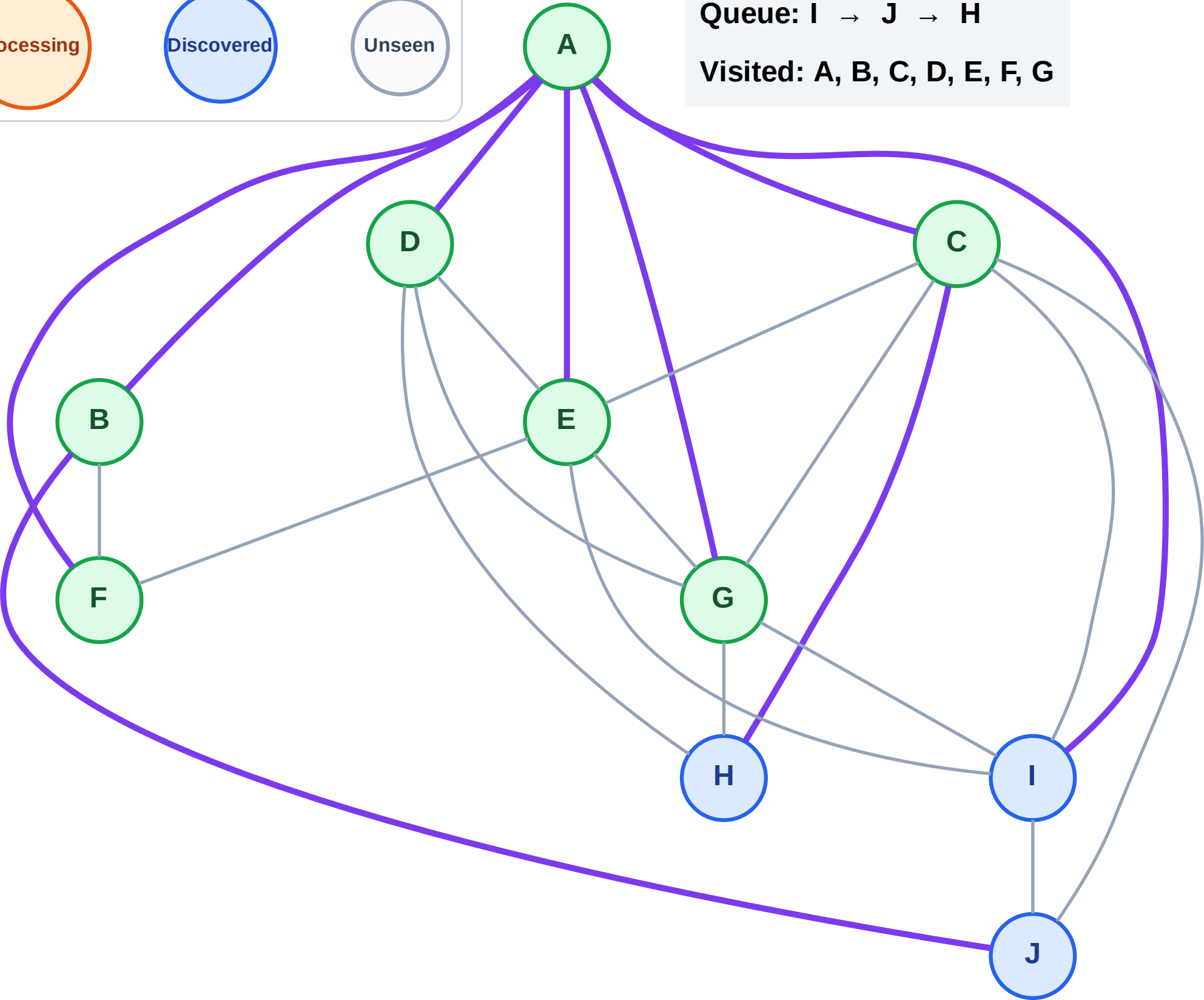
Processing

Discovered

Unseen

Queue: I → J → H

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 16: Process node I

Legend

Complete

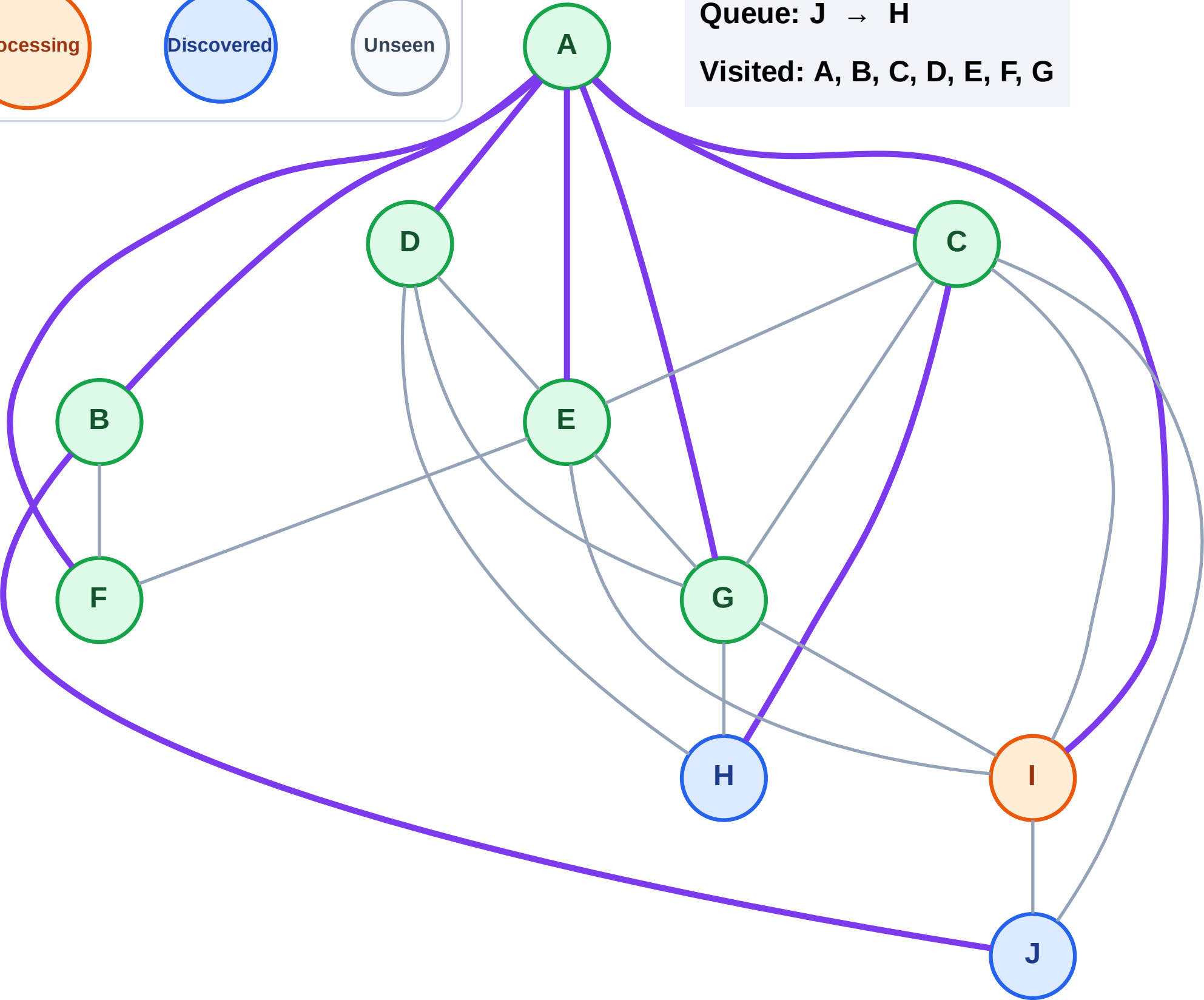
Processing

Discovered

Unseen

Queue: J → H

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 17: No new neighbors from I

Legend

Complete

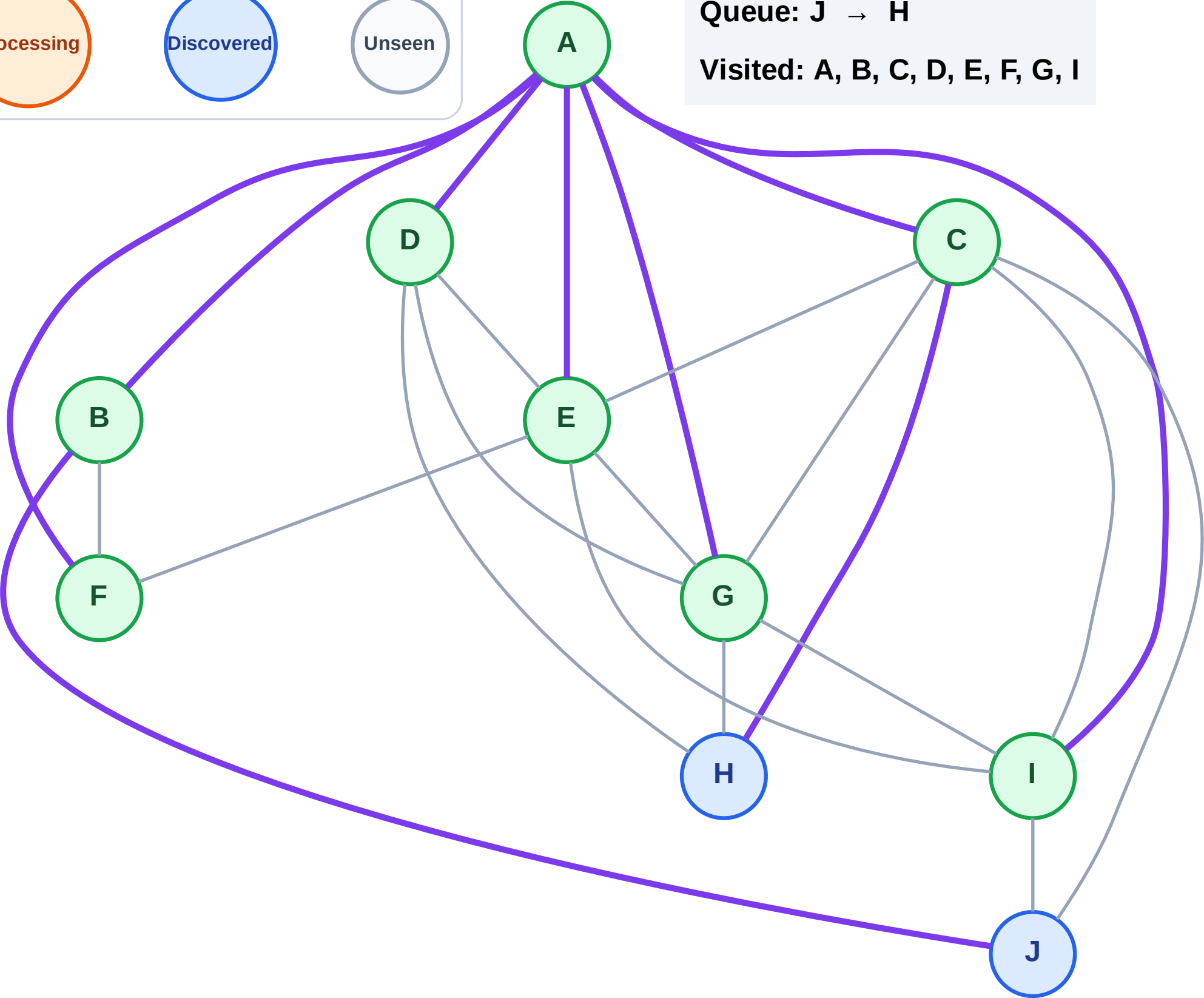
Processing

Discovered

Unseen

Queue: J → H

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node J

Legend

Complete

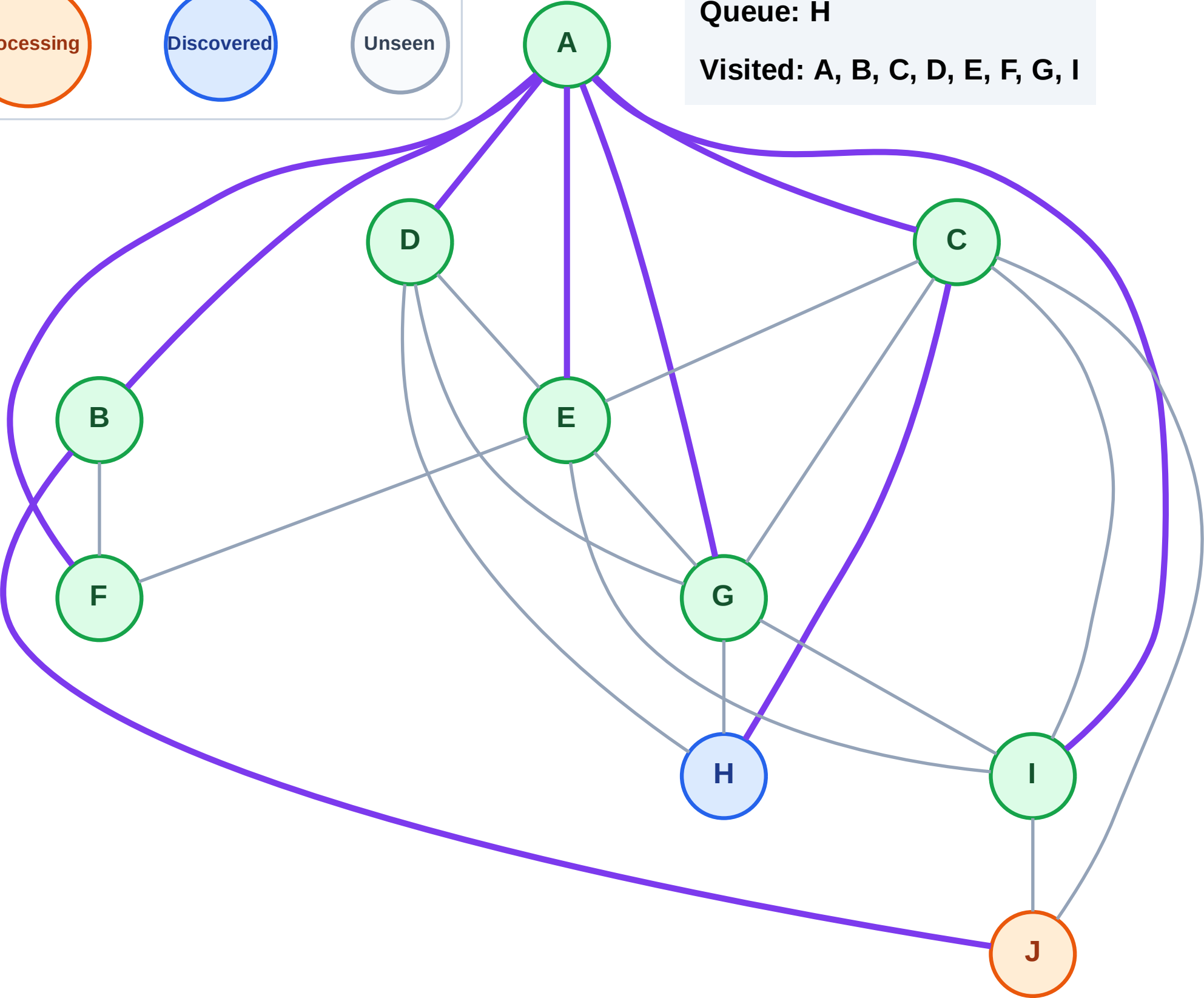
Processing

Discovered

Unseen

Queue: H

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 19: No new neighbors from J

Legend

Complete

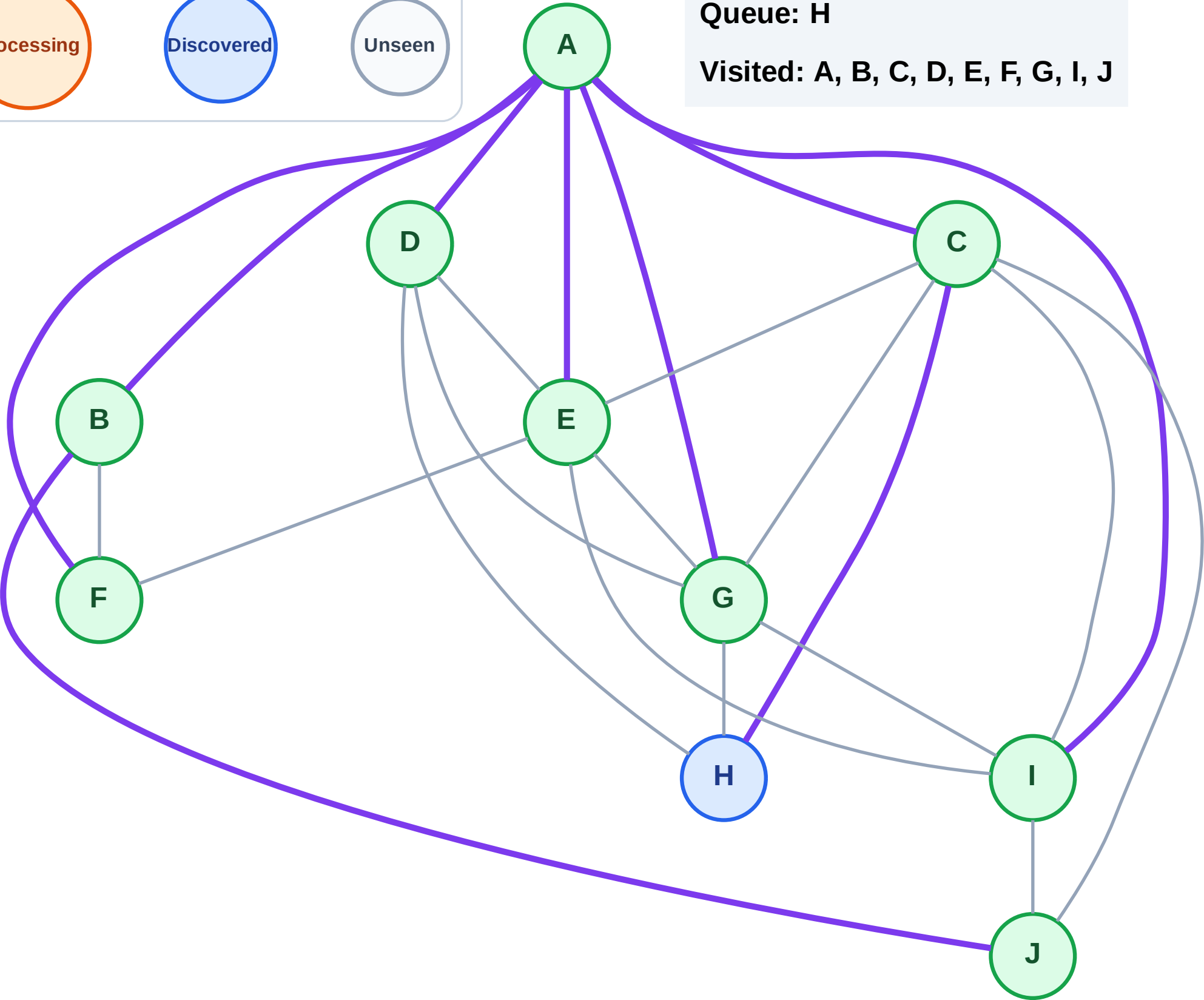
Processing

Discovered

Unseen

Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

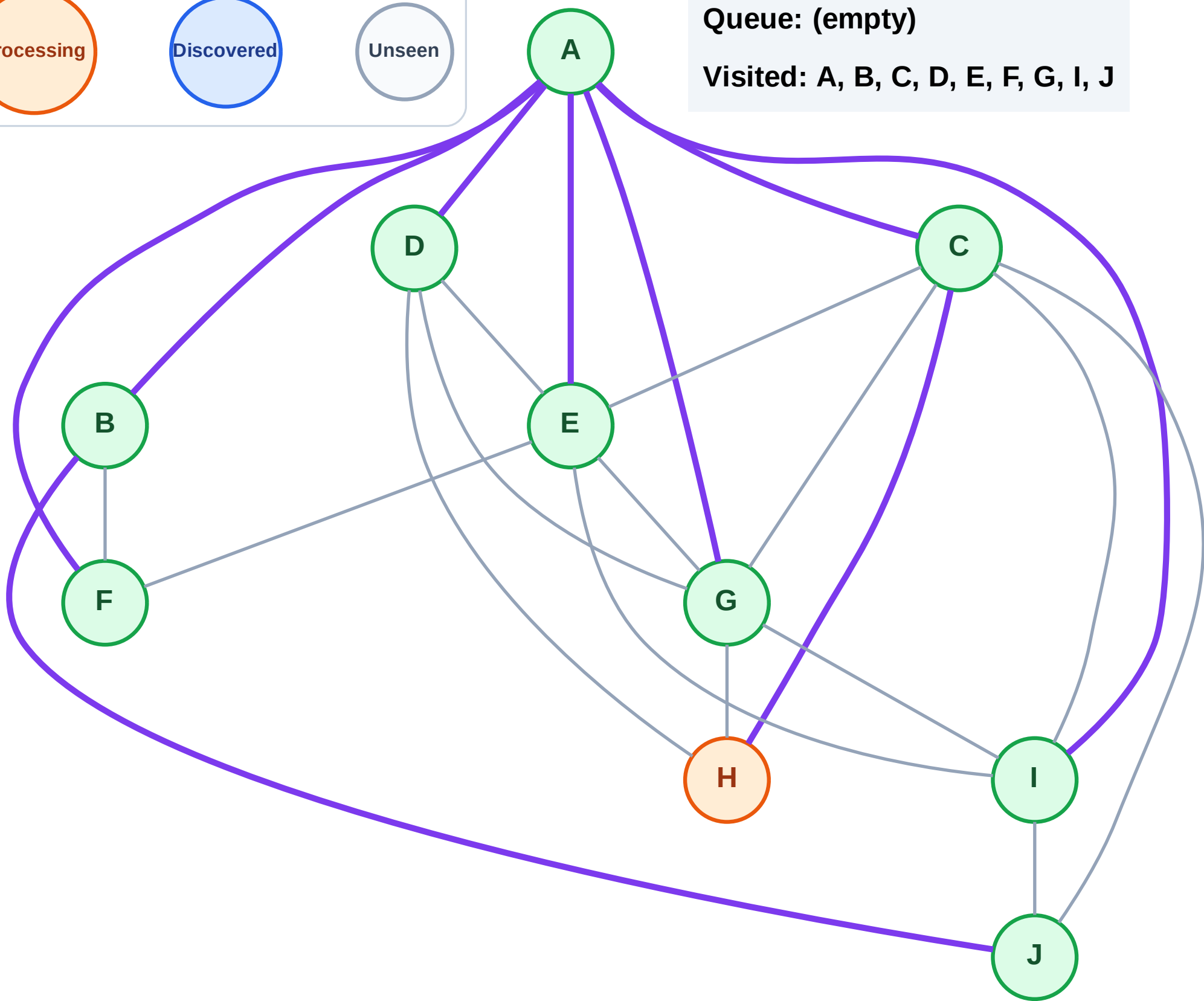
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

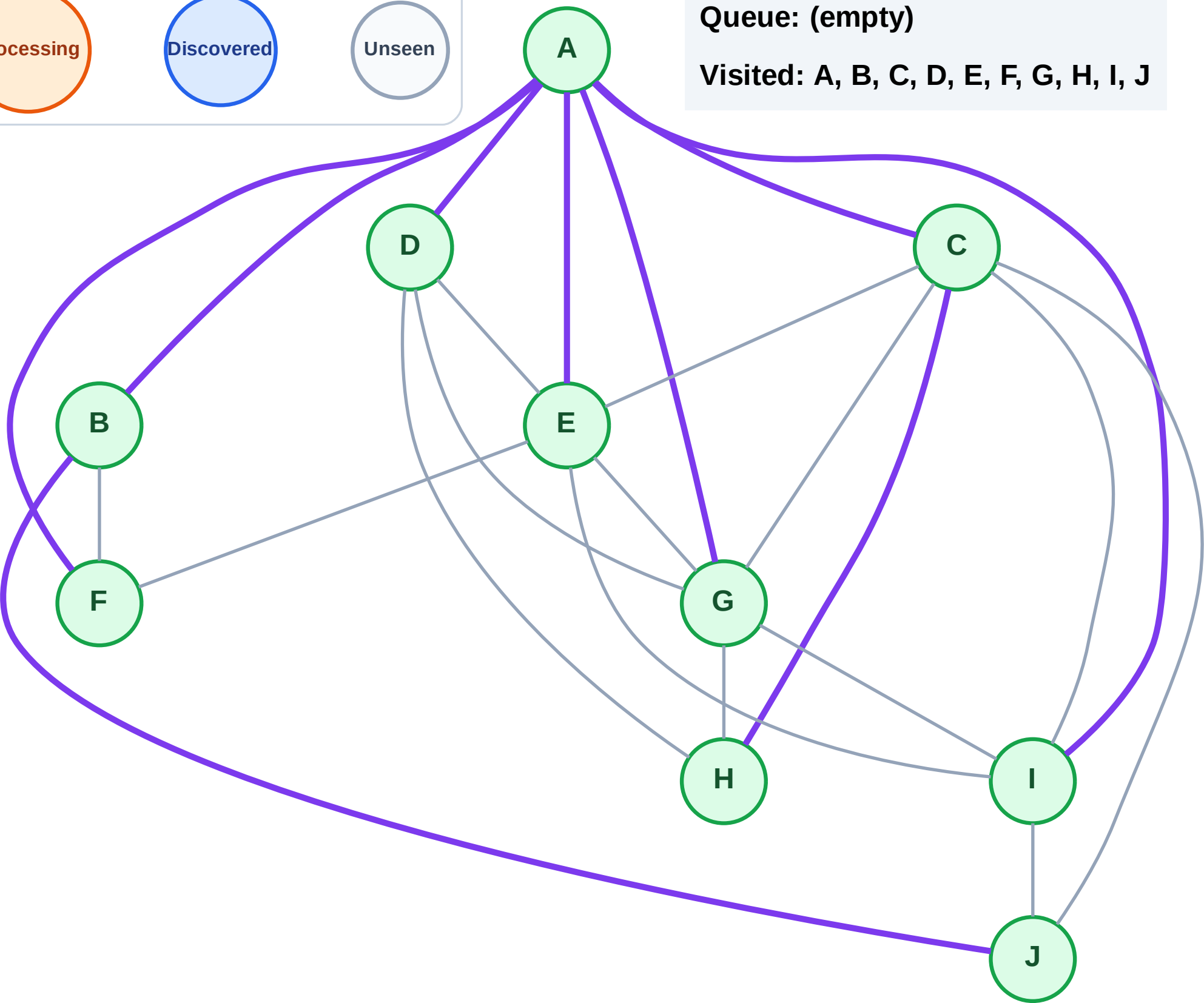
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

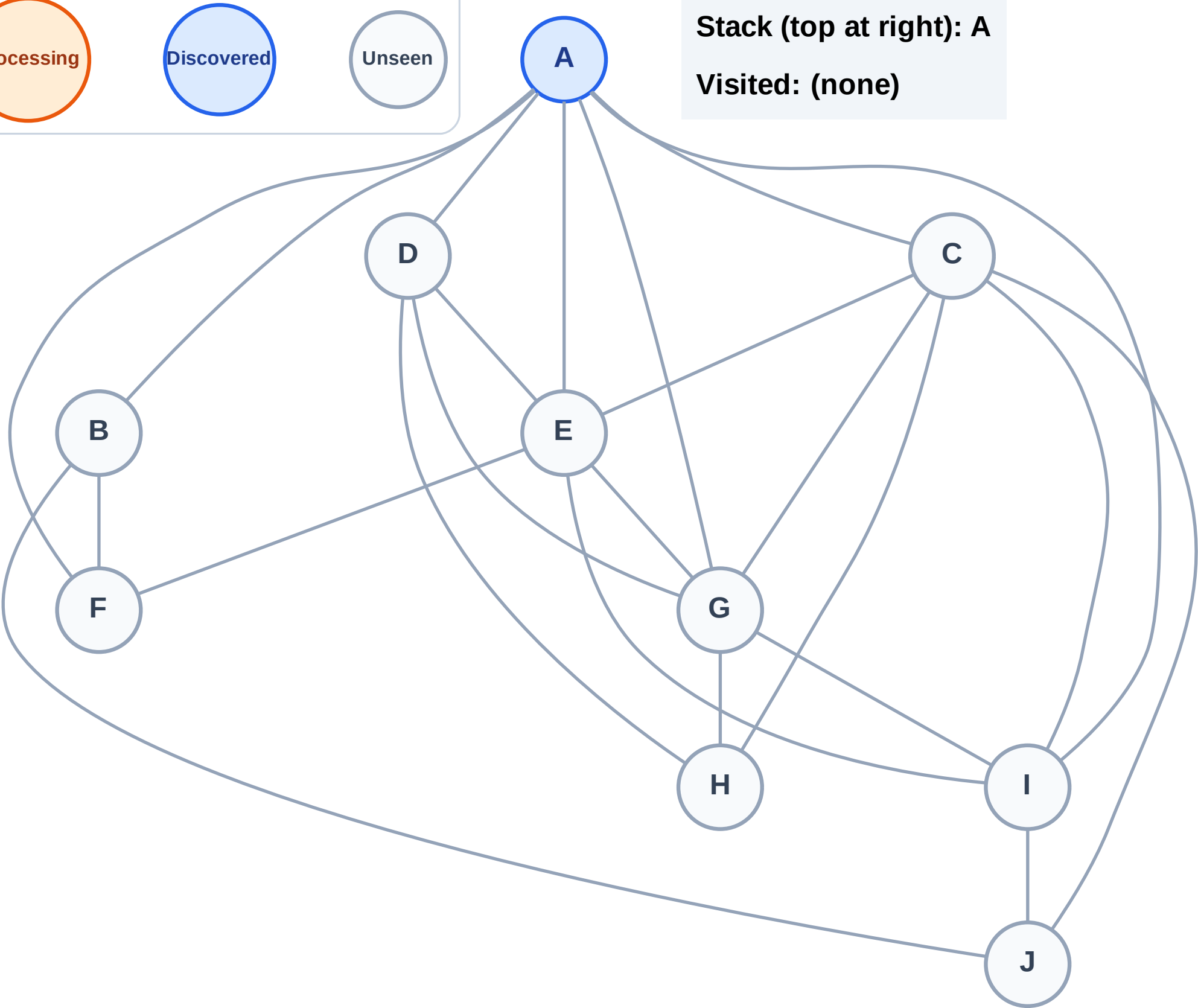
Complete

Processing

Discovered

Unseen

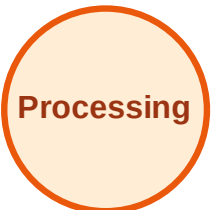
Stack (top at right): A
Visited: (none)



DFS Traversal

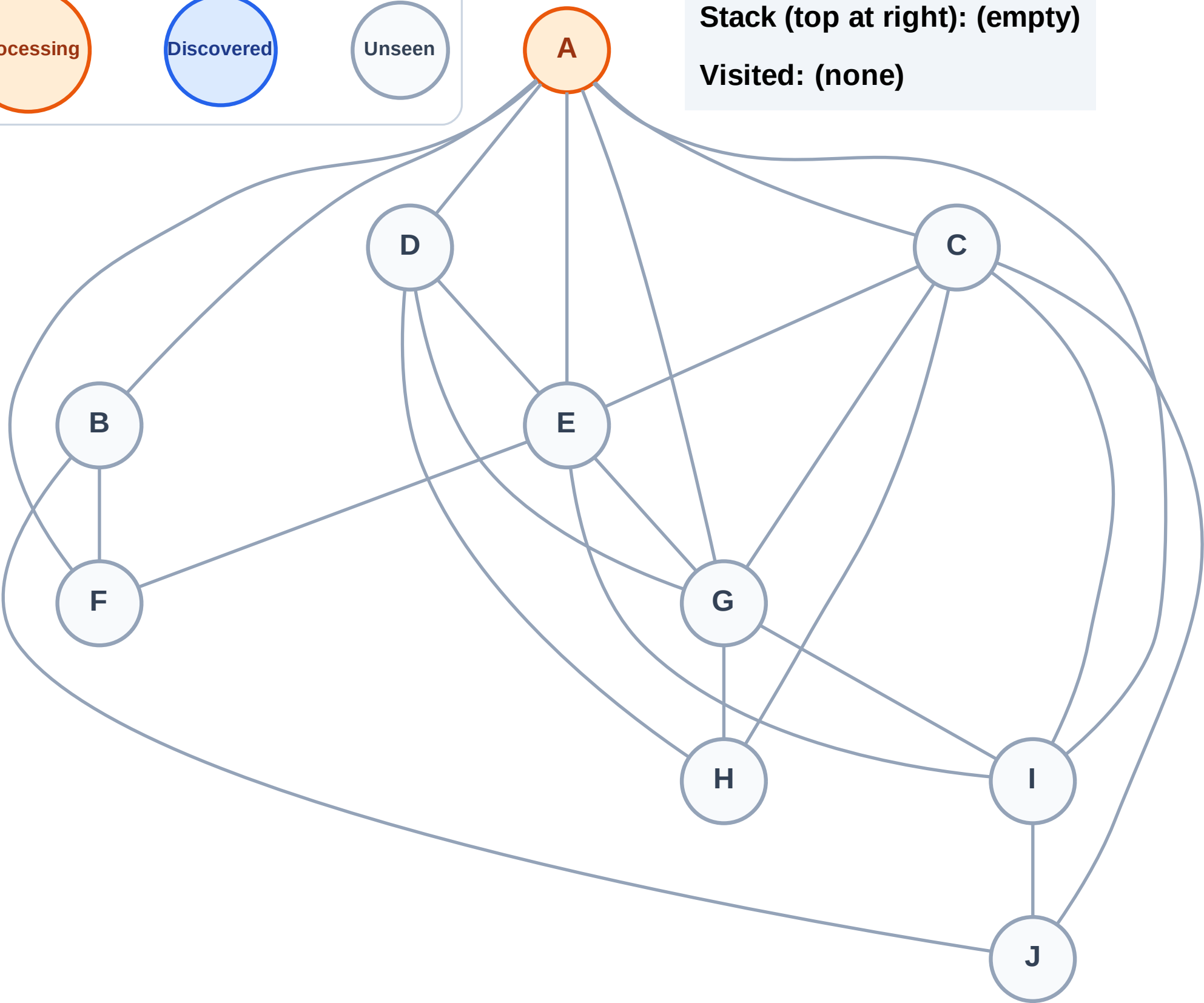
Step 2: Process node A

Legend



Stack (top at right): (empty)

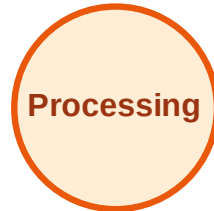
Visited: (none)



DFS Traversal

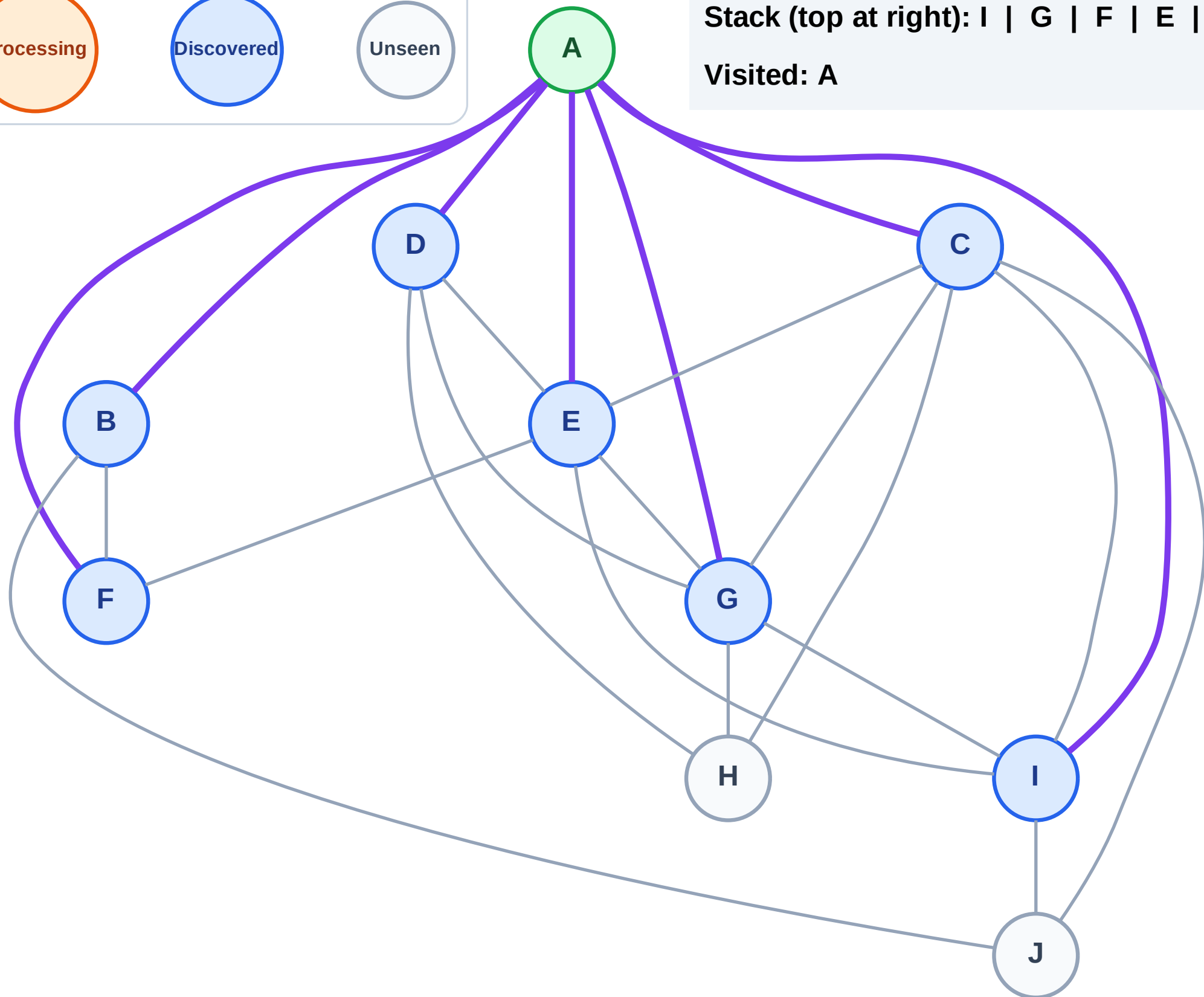
Step 3: Discover I, G, F, E, D, C, B from A

Legend



Stack (top at right): I | G | F | E | D | C | B

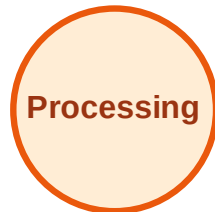
Visited: A



DFS Traversal

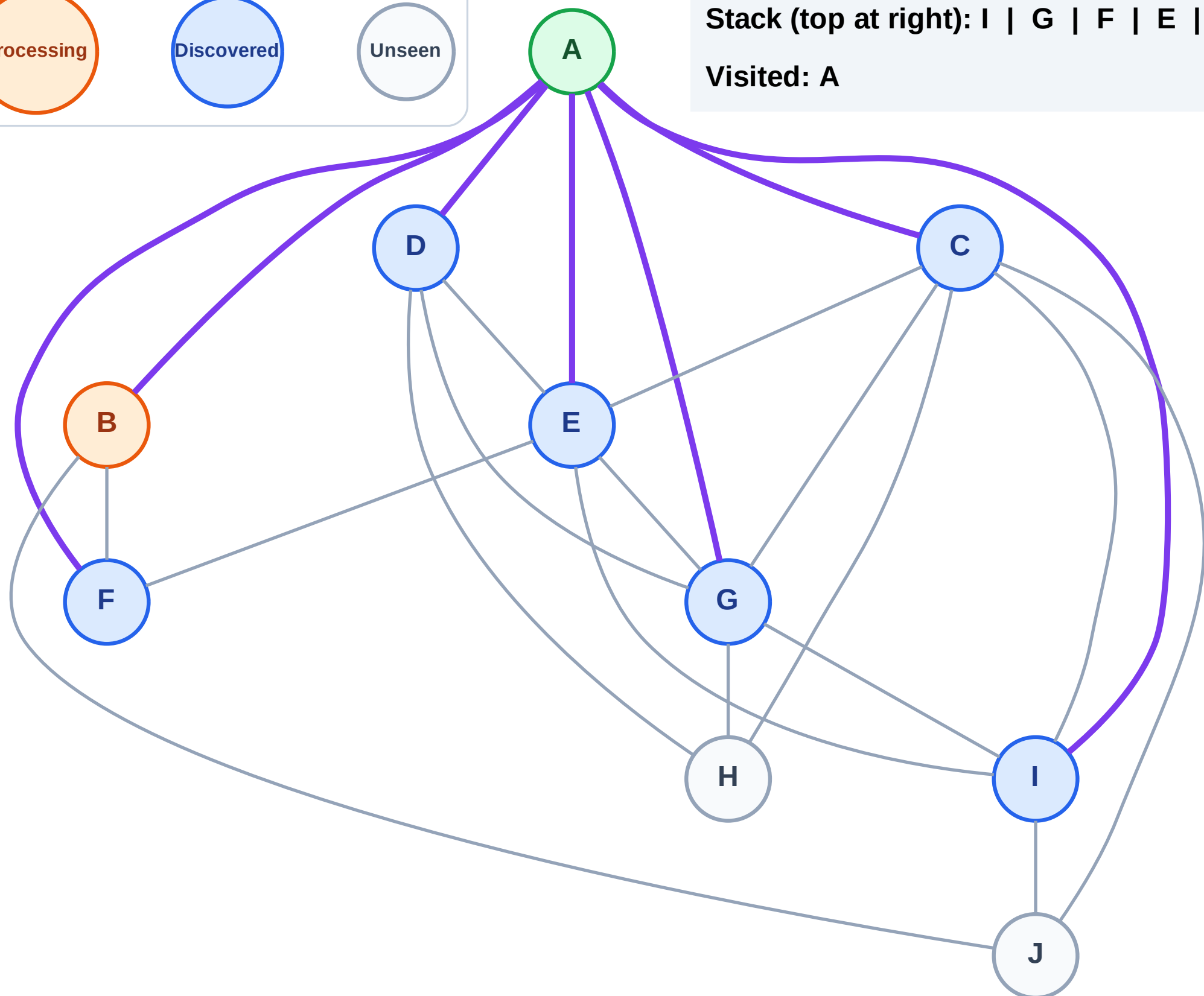
Step 4: Process node B

Legend



Stack (top at right): I | G | F | E | D | C

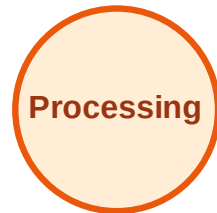
Visited: A



DFS Traversal

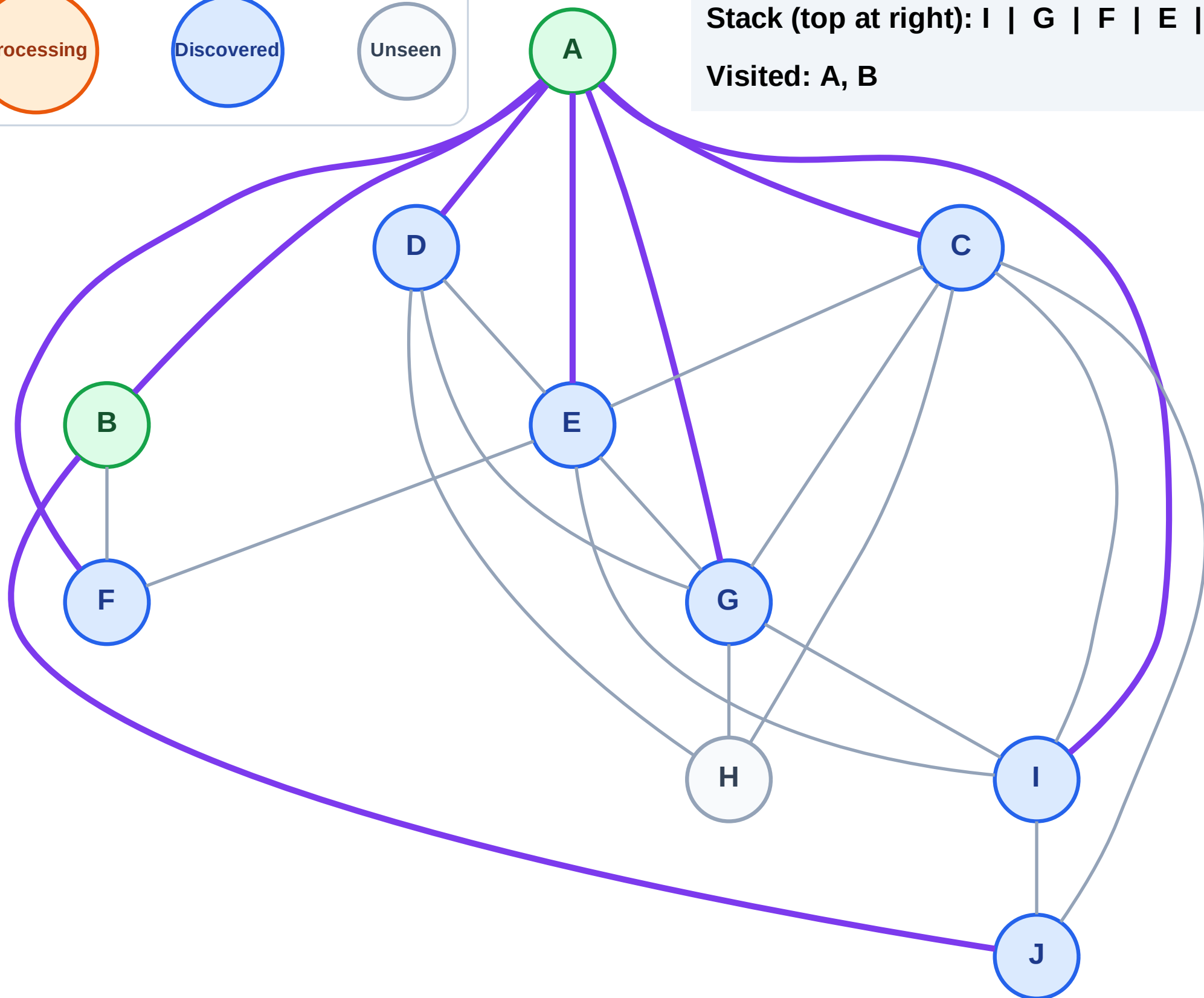
Step 5: Discover J from B

Legend



Stack (top at right): I | G | F | E | D | C | J

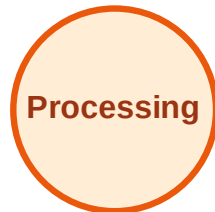
Visited: A, B



DFS Traversal

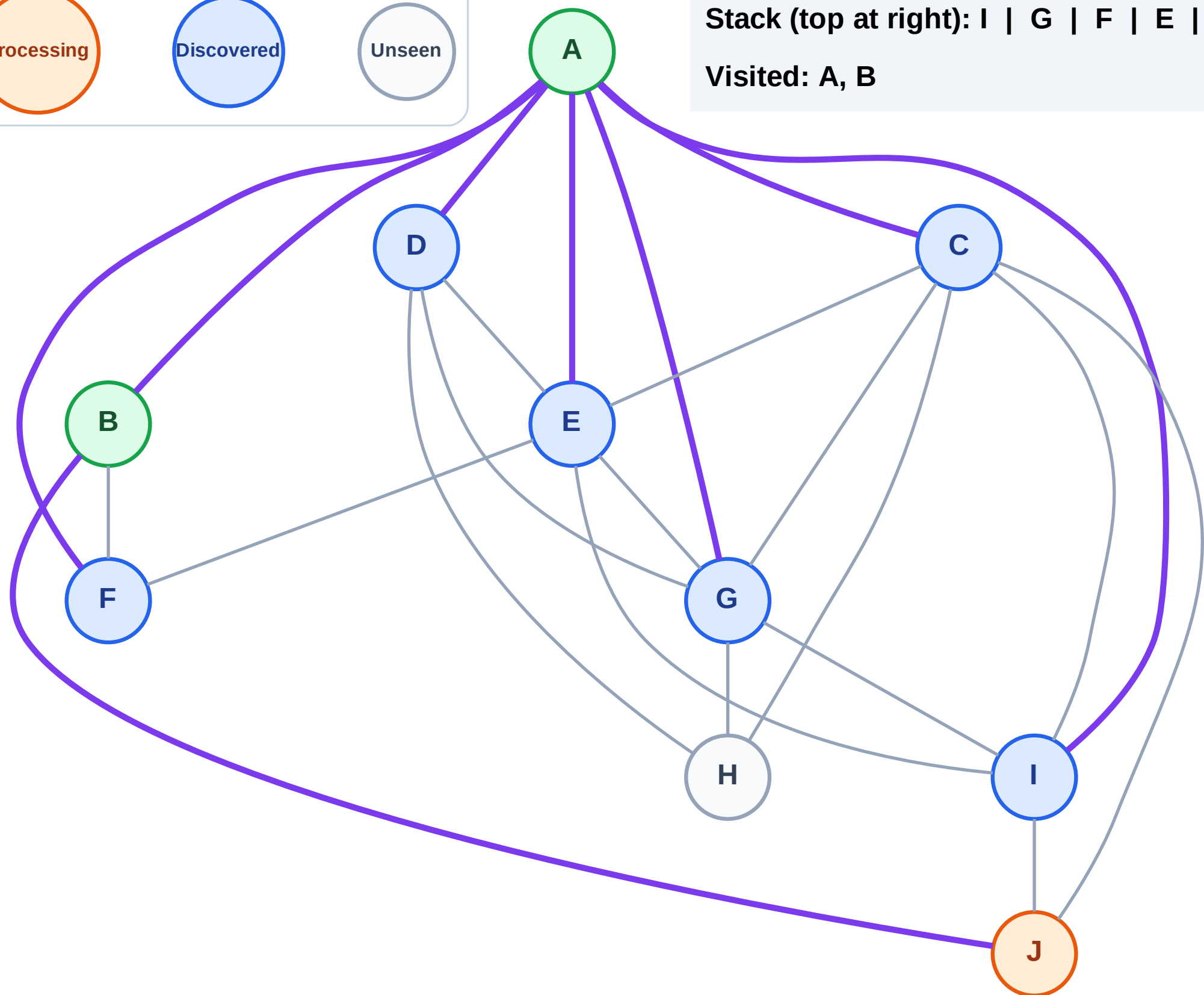
Step 6: Process node J

Legend



Stack (top at right): I | G | F | E | D | C

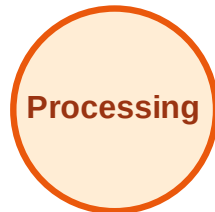
Visited: A, B



DFS Traversal

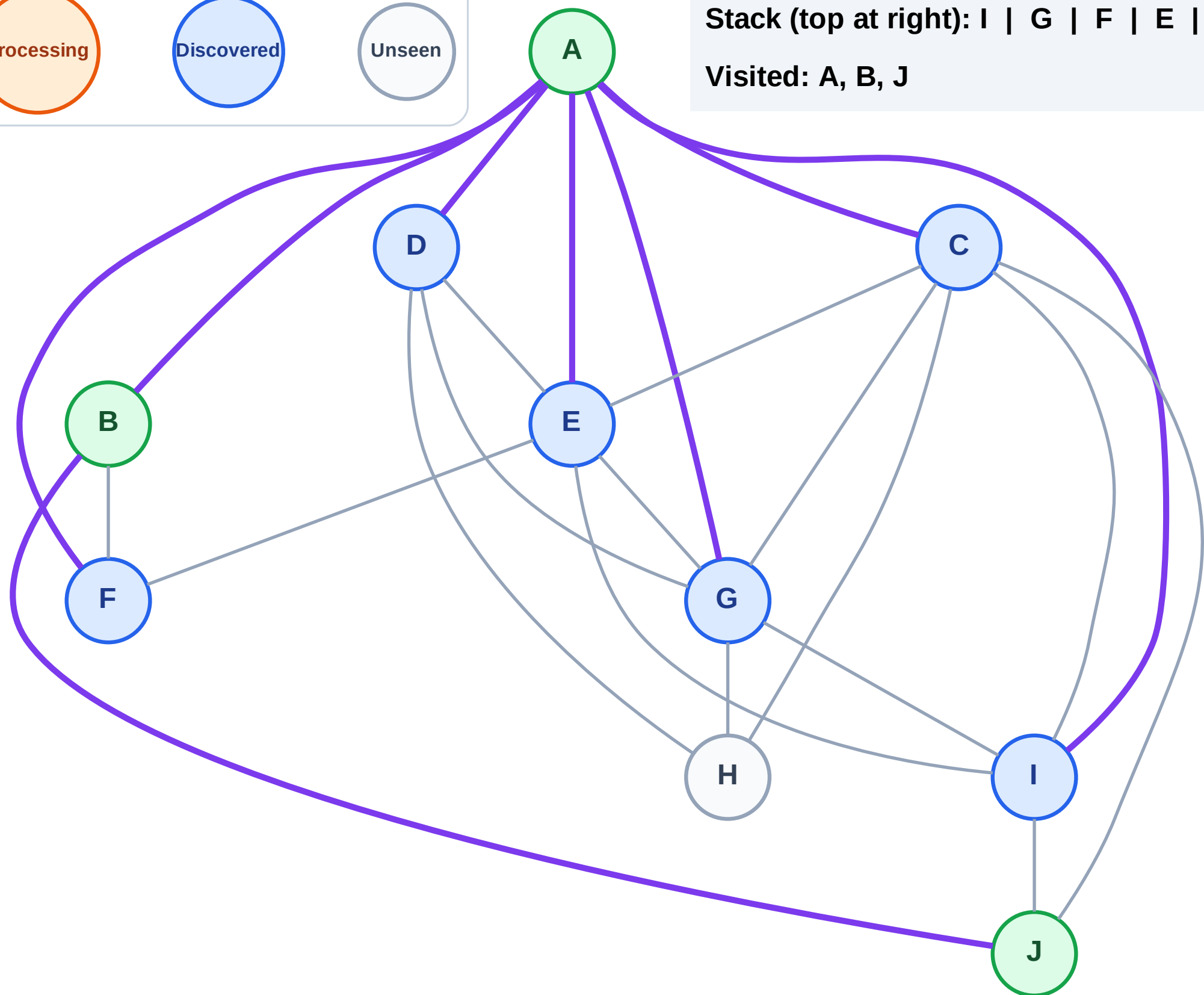
Step 7: No new neighbors from J

Legend



Stack (top at right): I | G | F | E | D | C

Visited: A, B, J



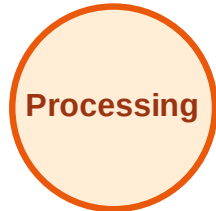
DFS Traversal

Step 8: Process node C

Legend



Complete



Processing



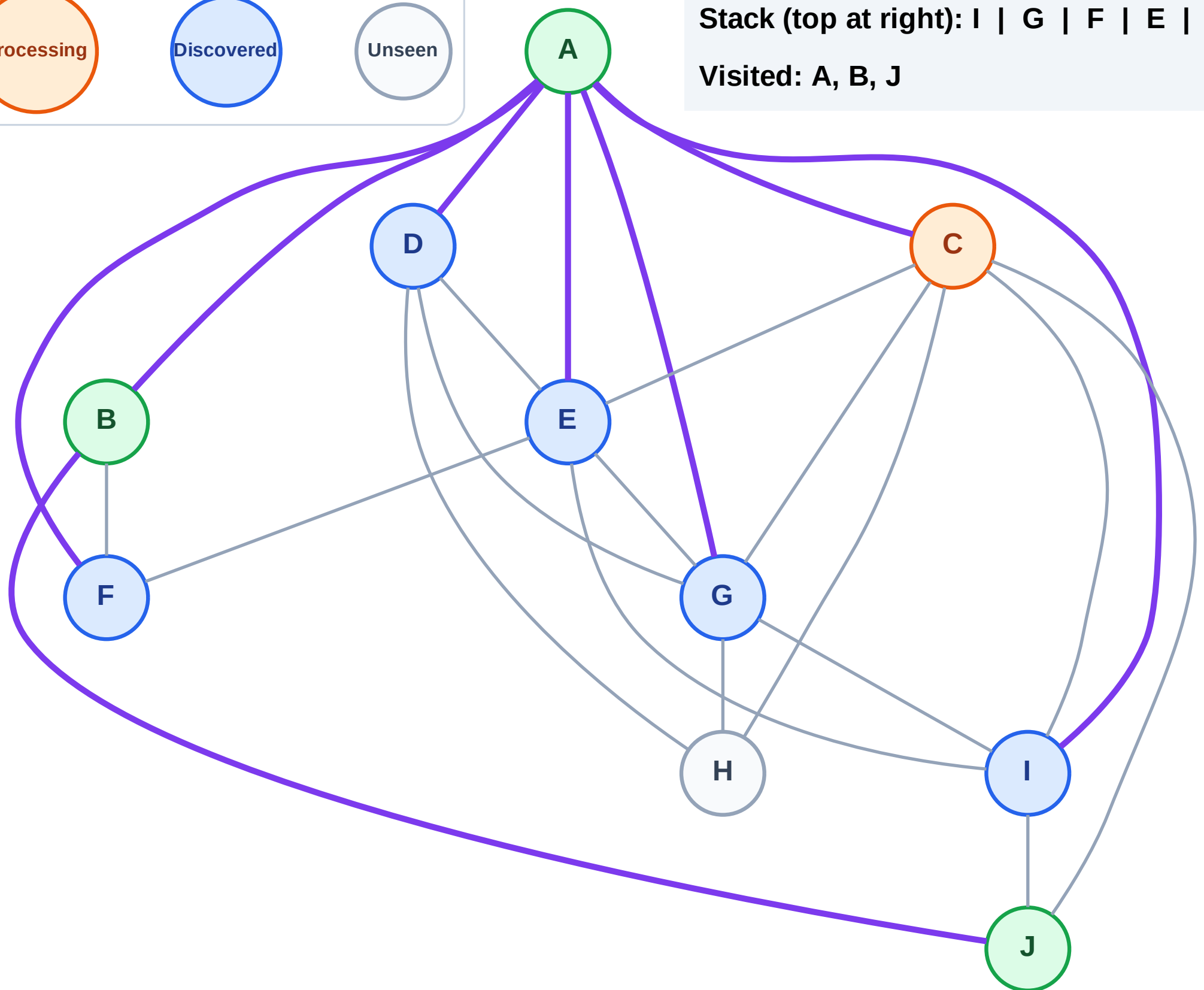
Discovered



Unseen

Stack (top at right): I | G | F | E | D

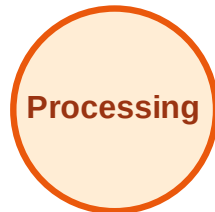
Visited: A, B, J



DFS Traversal

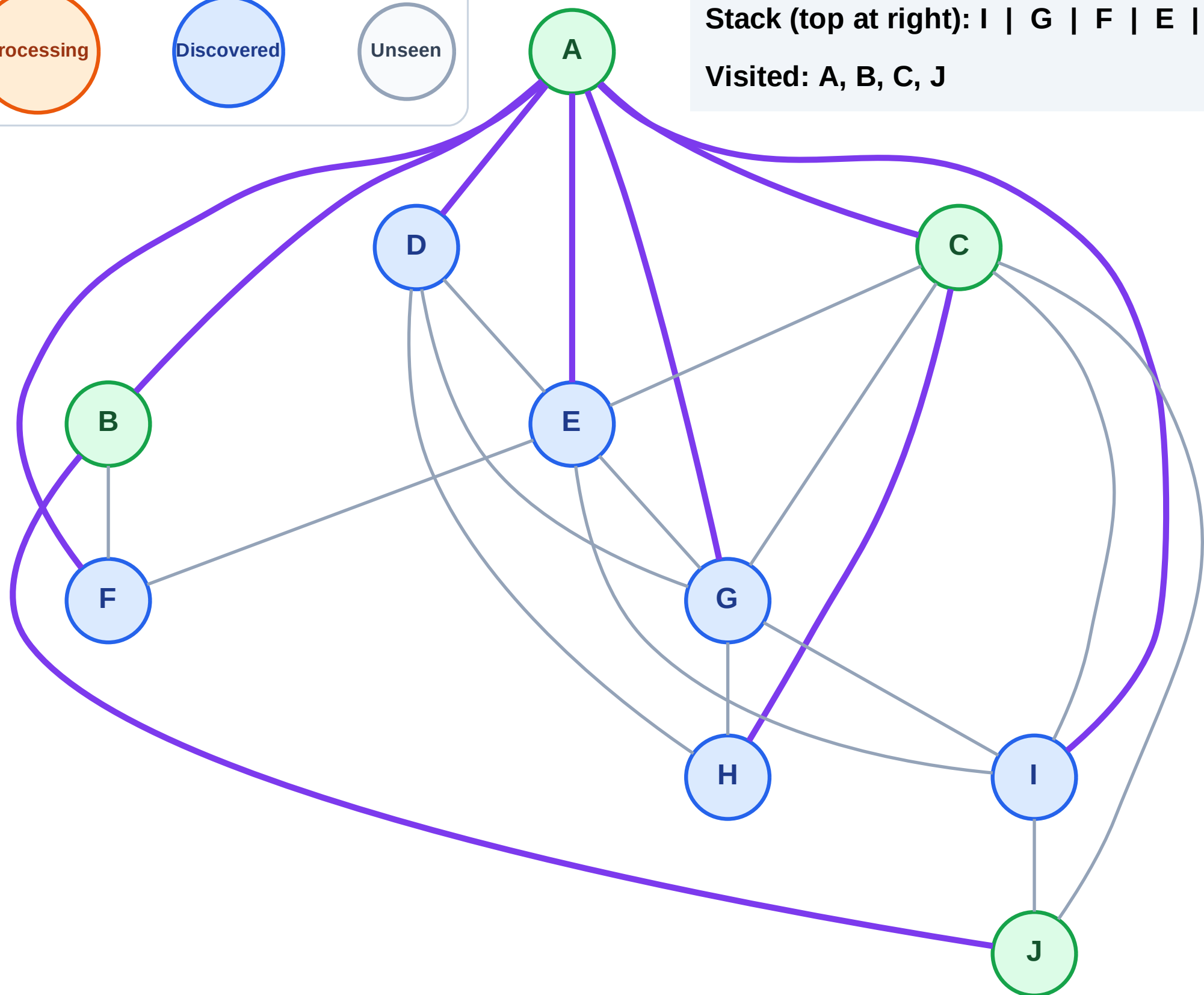
Step 9: Discover H from C

Legend



Stack (top at right): I | G | F | E | D | H

Visited: A, B, C, J



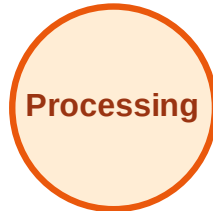
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



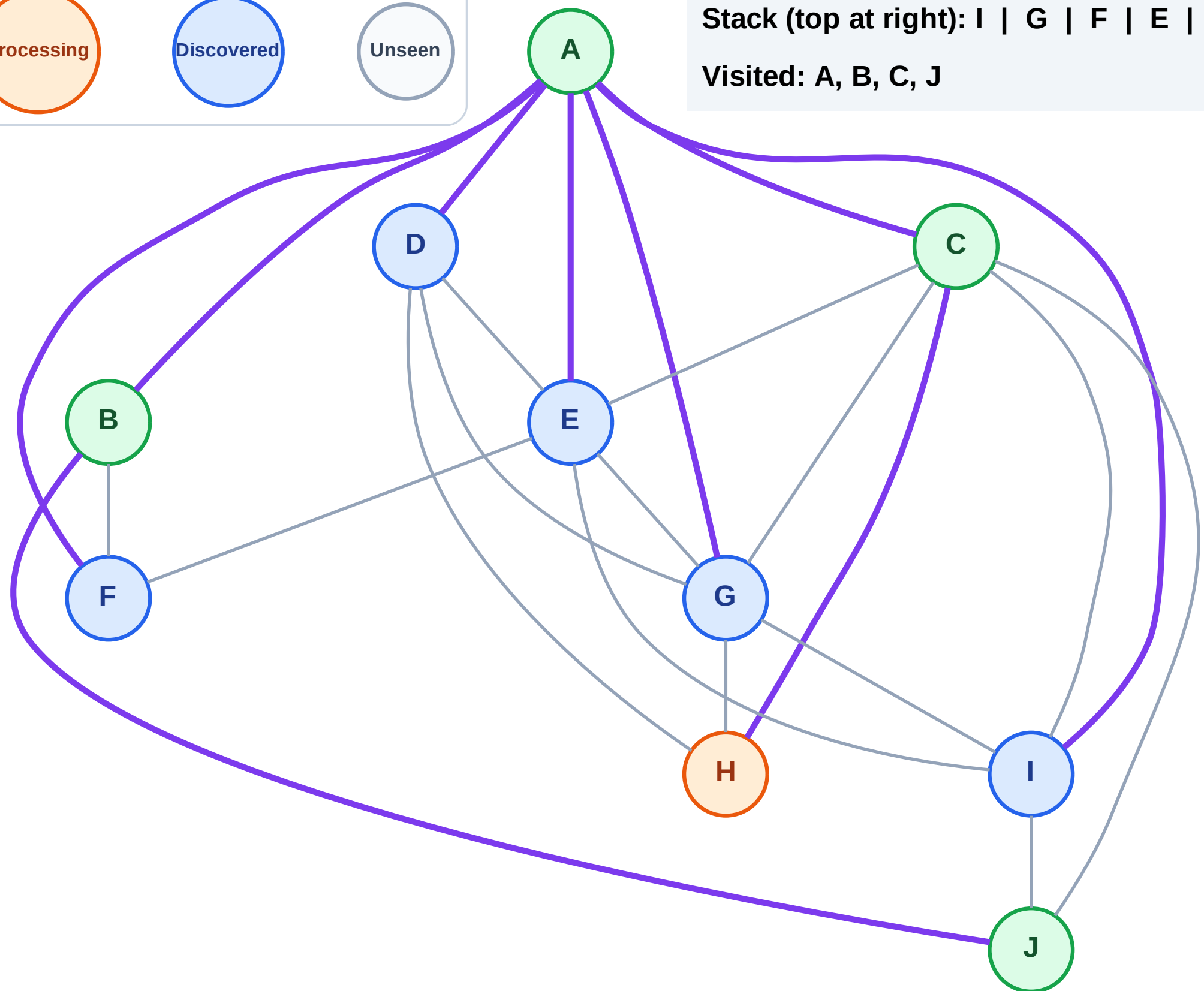
Discovered



Unseen

Stack (top at right): I | G | F | E | D

Visited: A, B, C, J



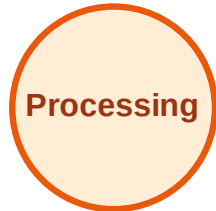
DFS Traversal

Step 11: No new neighbors from H

Legend



Complete



Processing



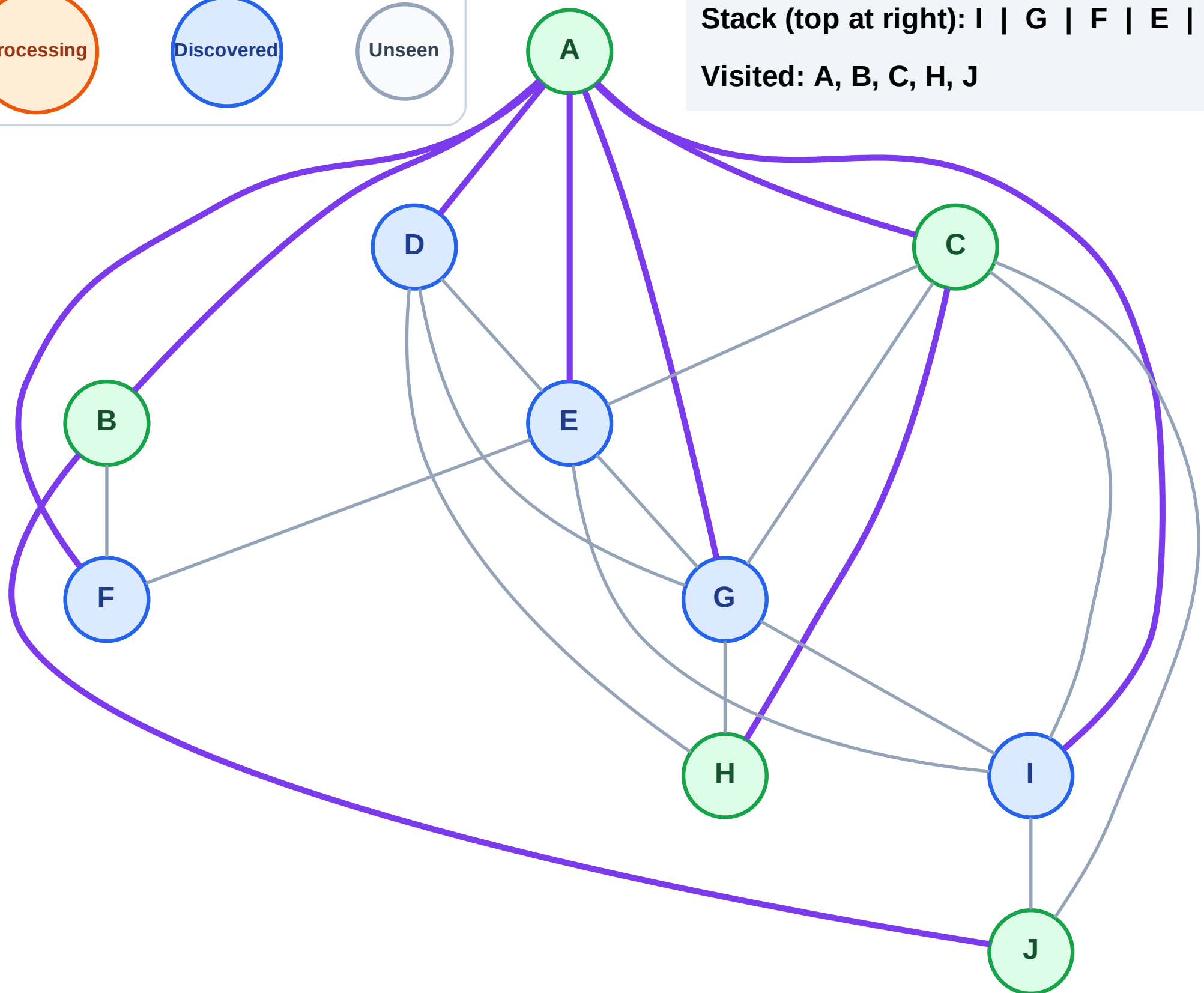
Discovered



Unseen

Stack (top at right): I | G | F | E | D

Visited: A, B, C, H, J



DFS Traversal

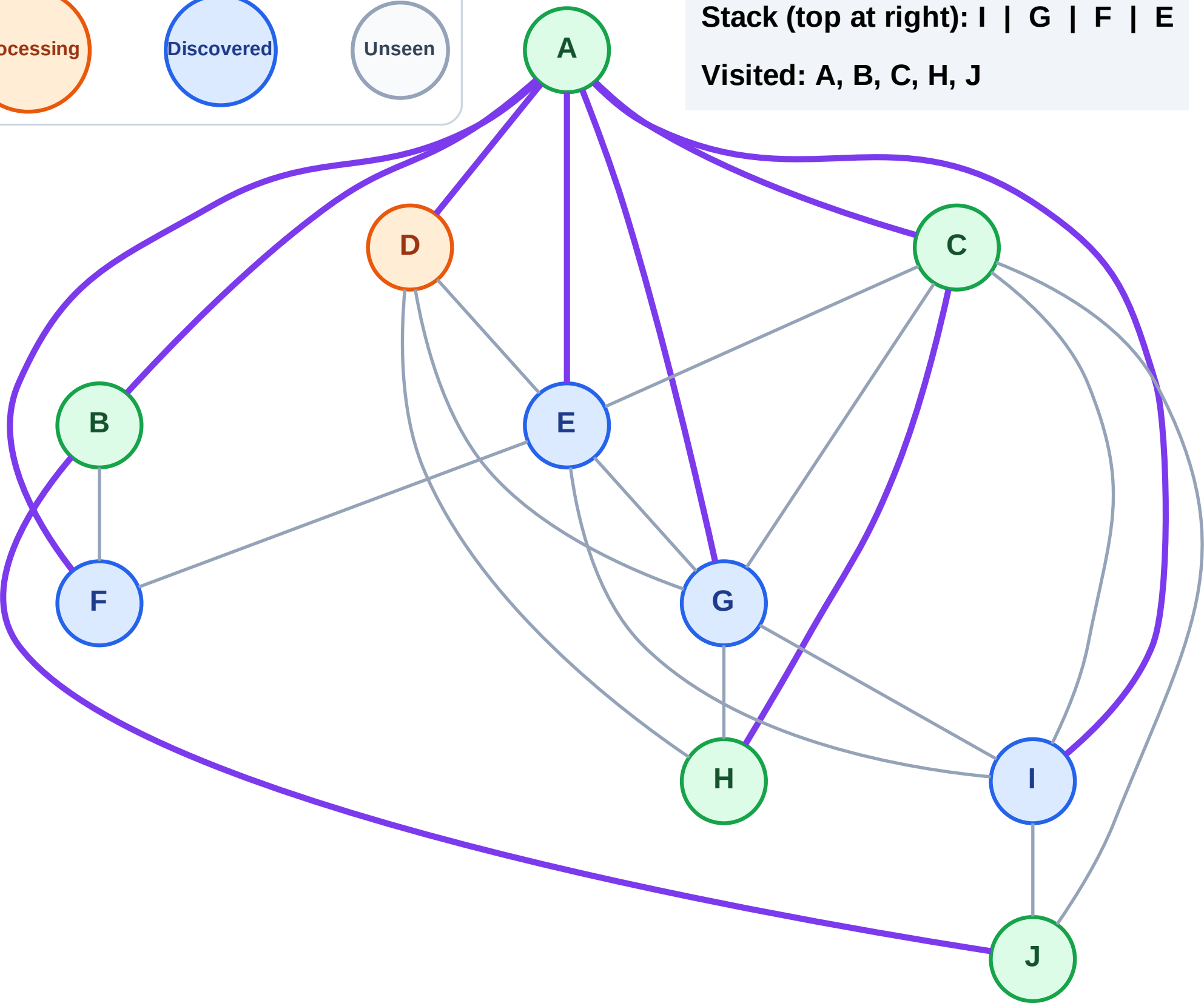
Step 12: Process node D

Legend



Stack (top at right): I | G | F | E

Visited: A, B, C, H, J



DFS Traversal

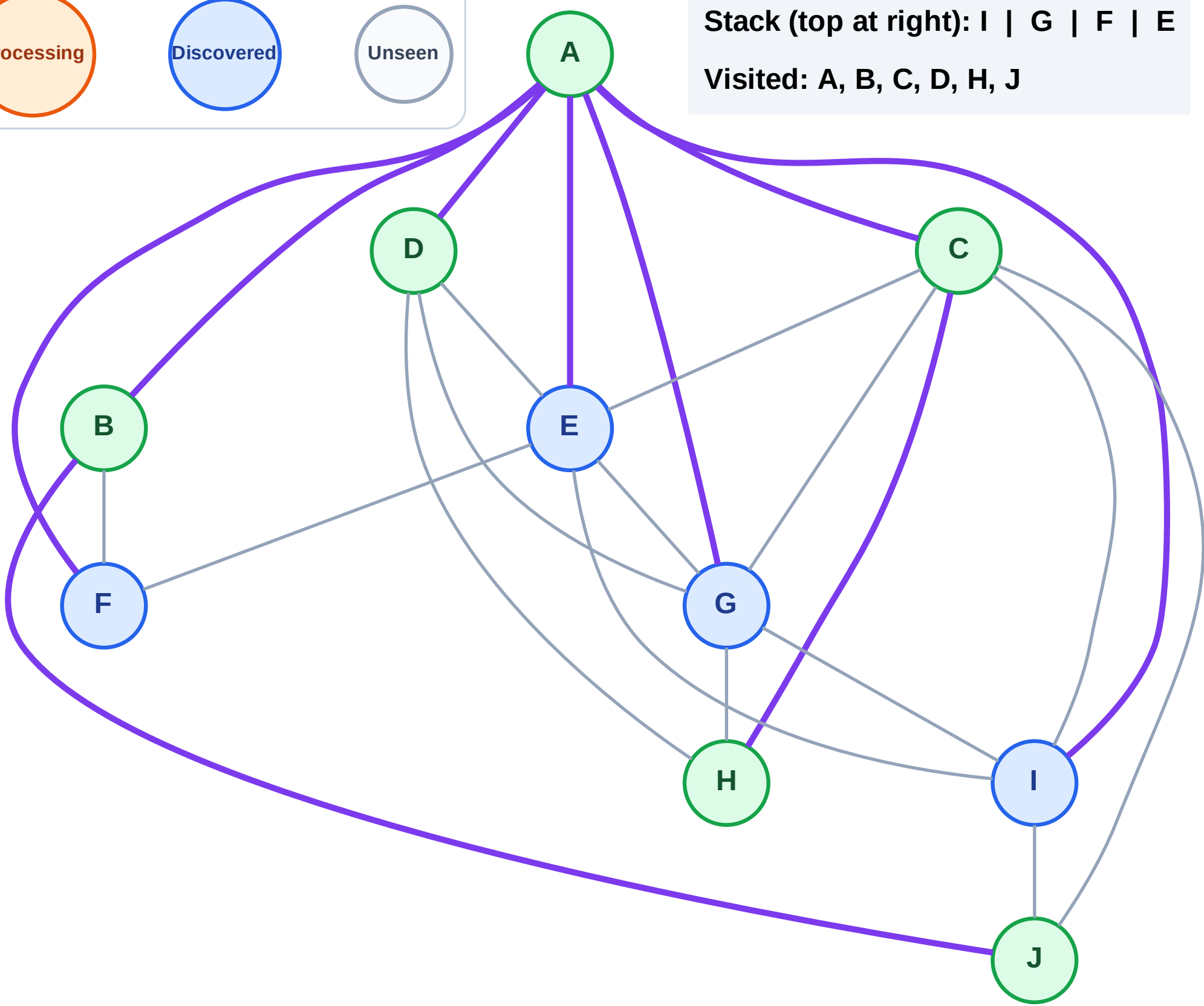
Step 13: No new neighbors from D

Legend



Stack (top at right): I | G | F | E

Visited: A, B, C, D, H, J



DFS Traversal

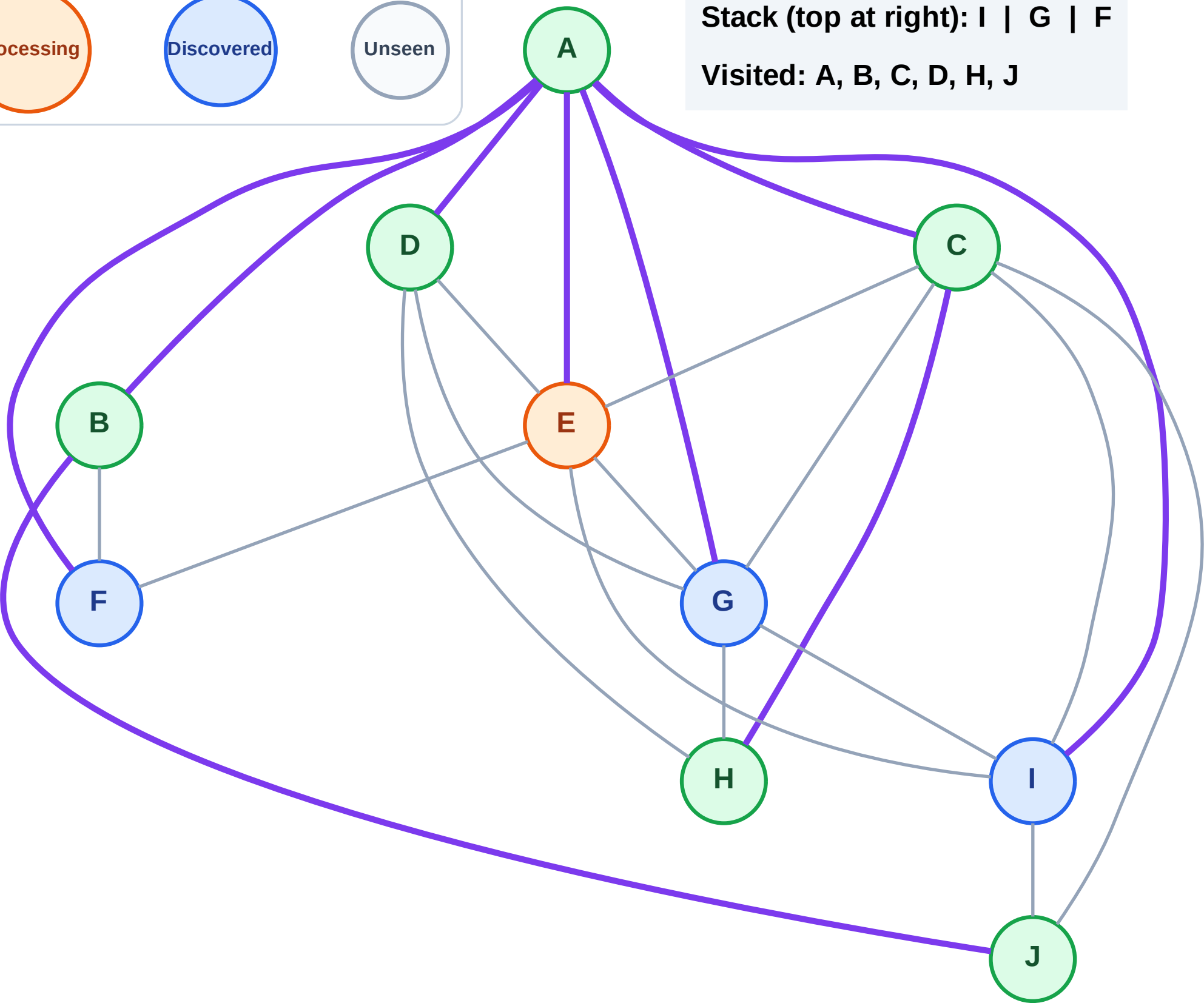
Step 14: Process node E

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, H, J



DFS Traversal

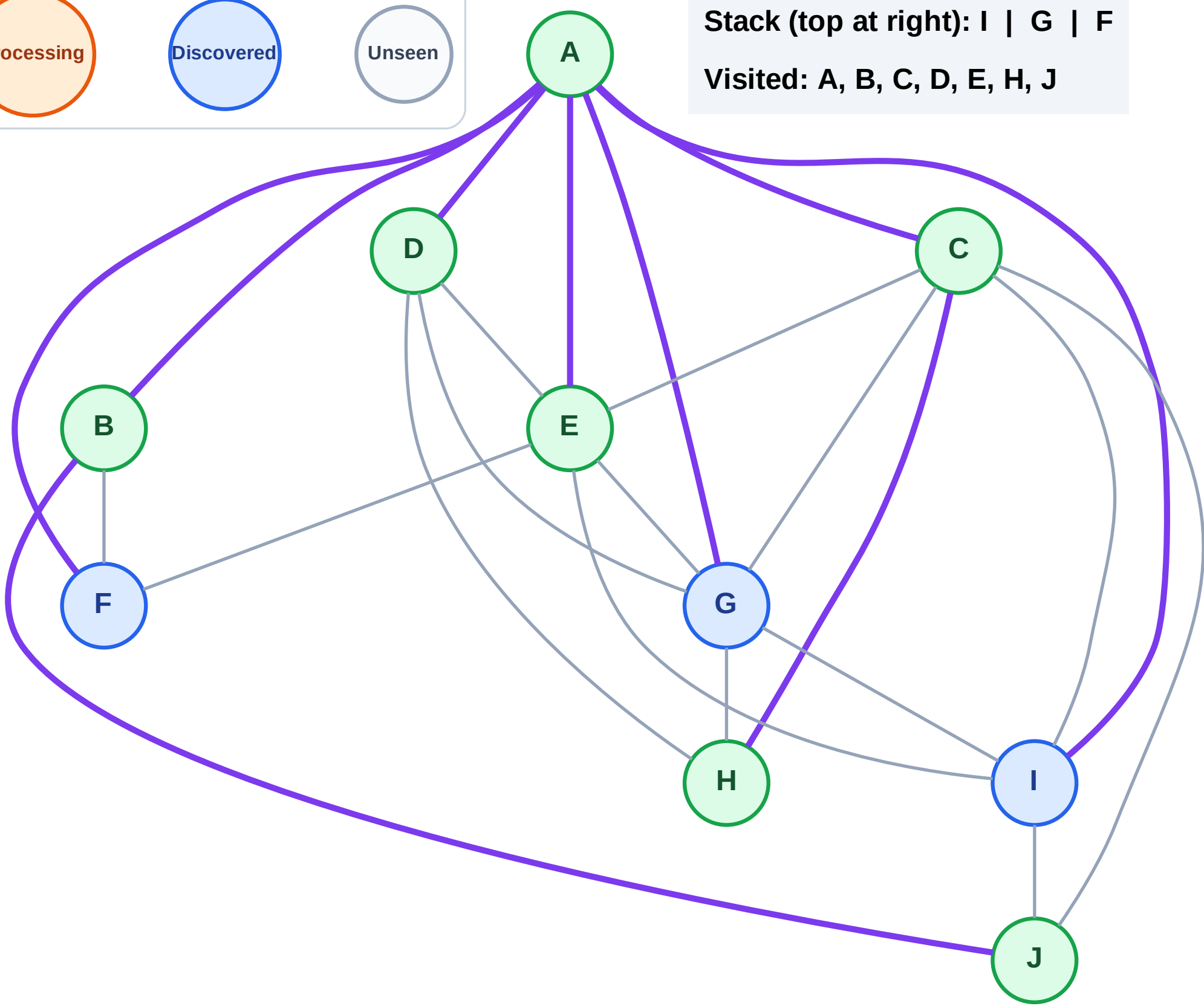
Step 15: No new neighbors from E

Legend



Stack (top at right): I | G | F

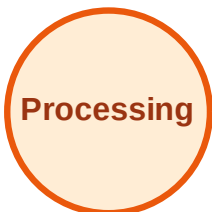
Visited: A, B, C, D, E, H, J



DFS Traversal

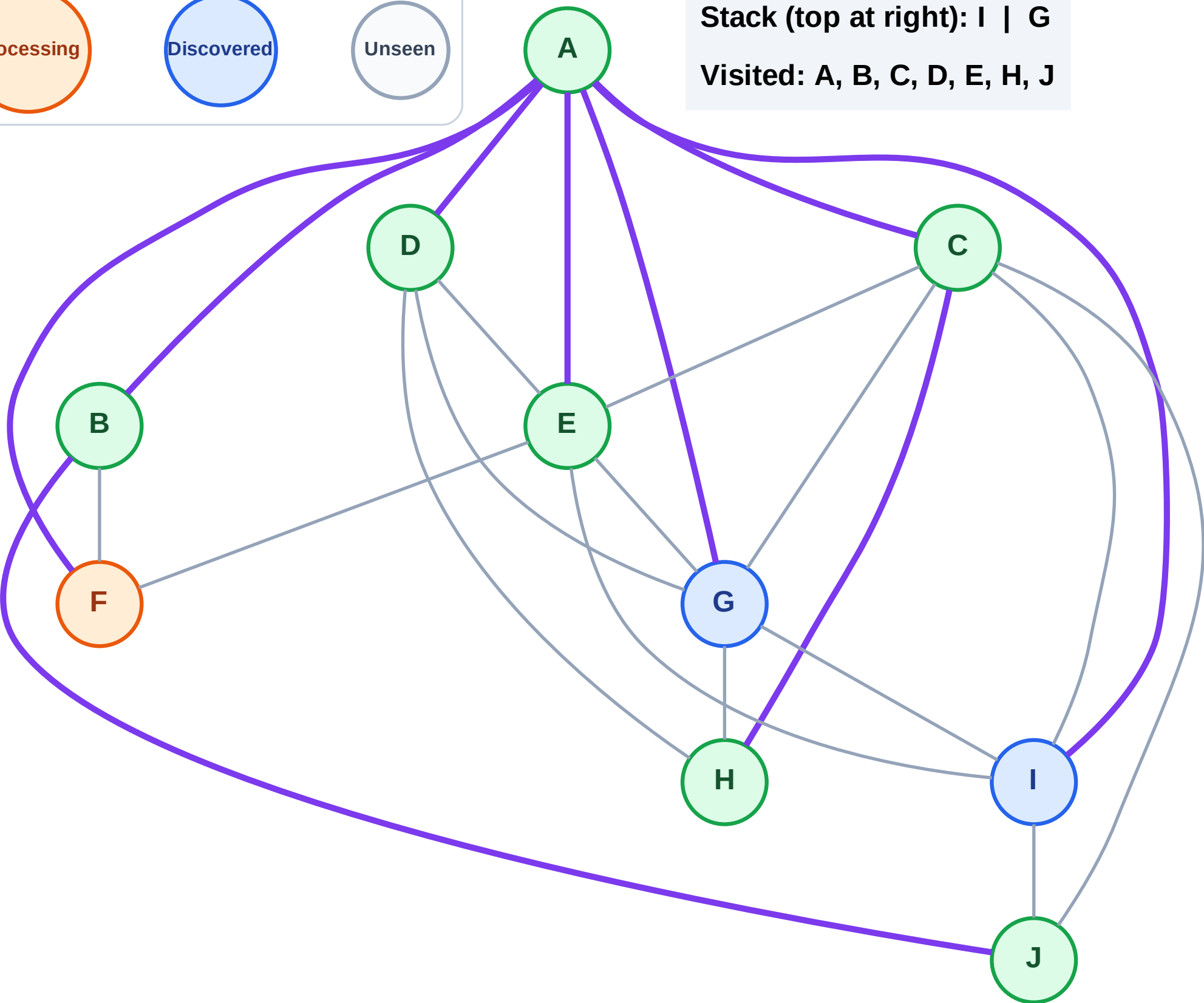
Step 16: Process node F

Legend



Stack (top at right): I | G

Visited: A, B, C, D, E, H, J



DFS Traversal

Step 17: No new neighbors from F

Legend

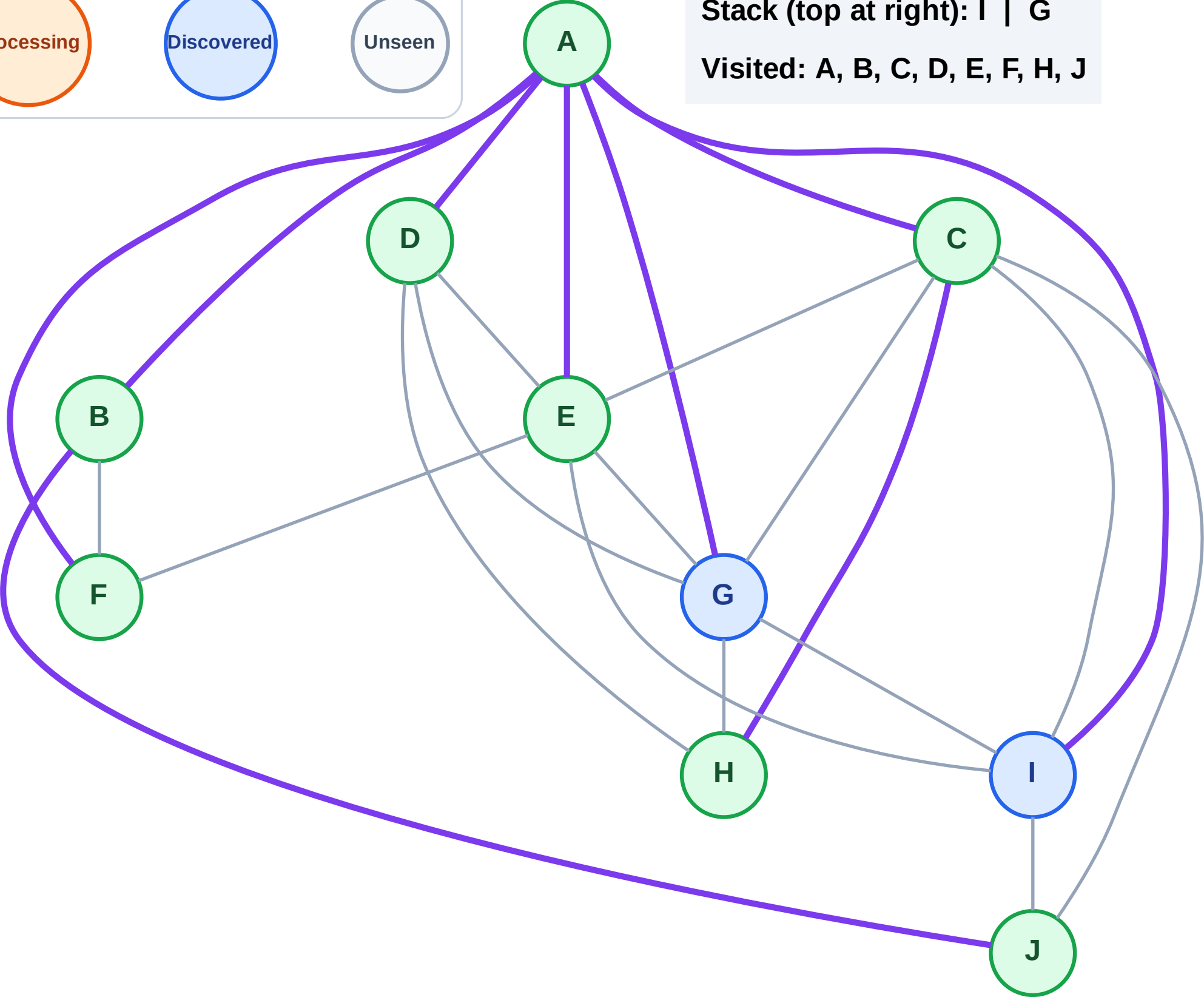
Complete

Processing

Discovered

Unseen

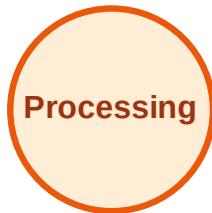
Stack (top at right): I | G
Visited: A, B, C, D, E, F, H, J



DFS Traversal

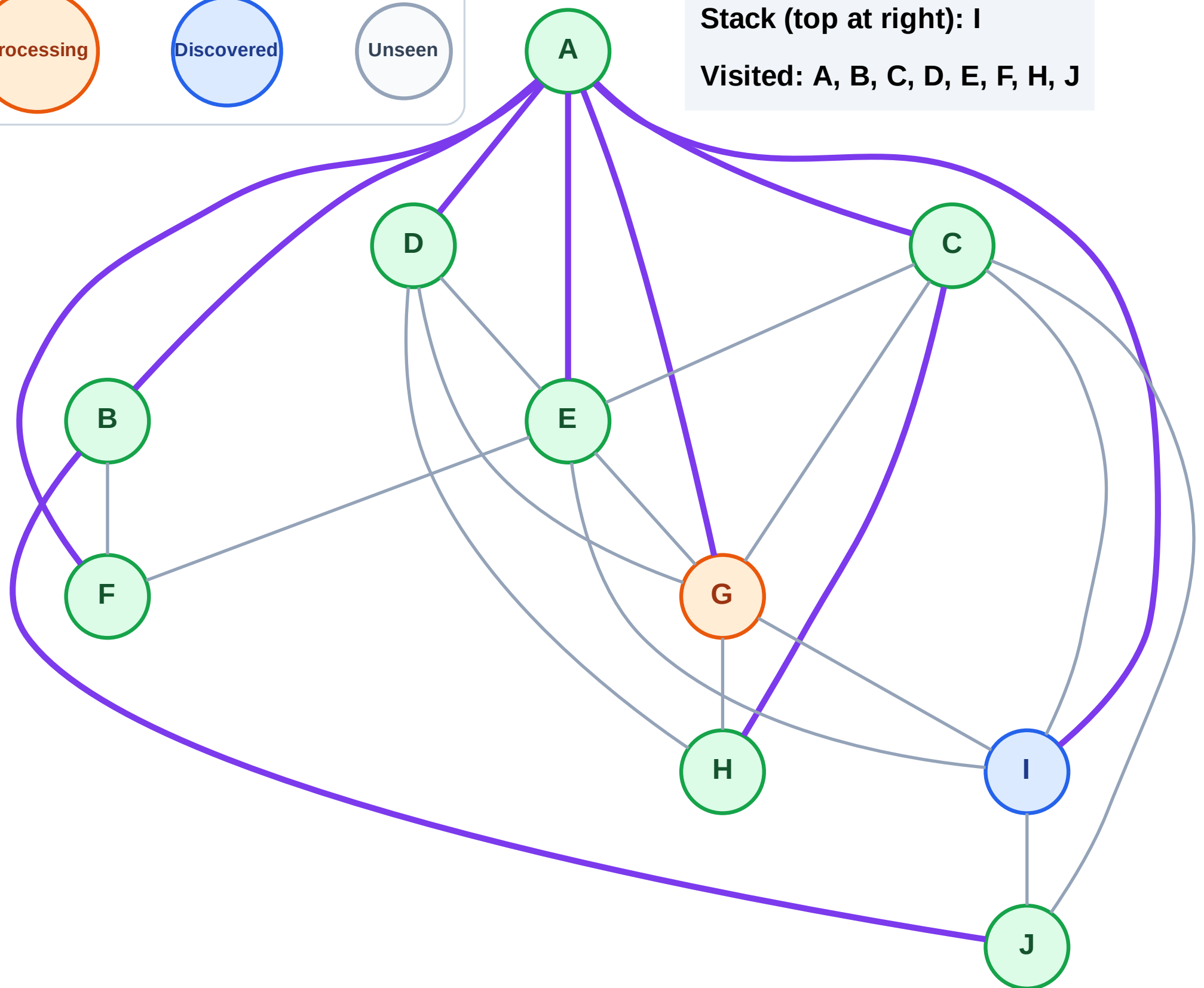
Step 18: Process node G

Legend



Stack (top at right): I

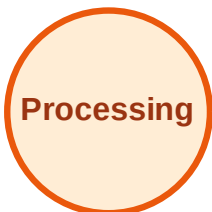
Visited: A, B, C, D, E, F, H, J



DFS Traversal

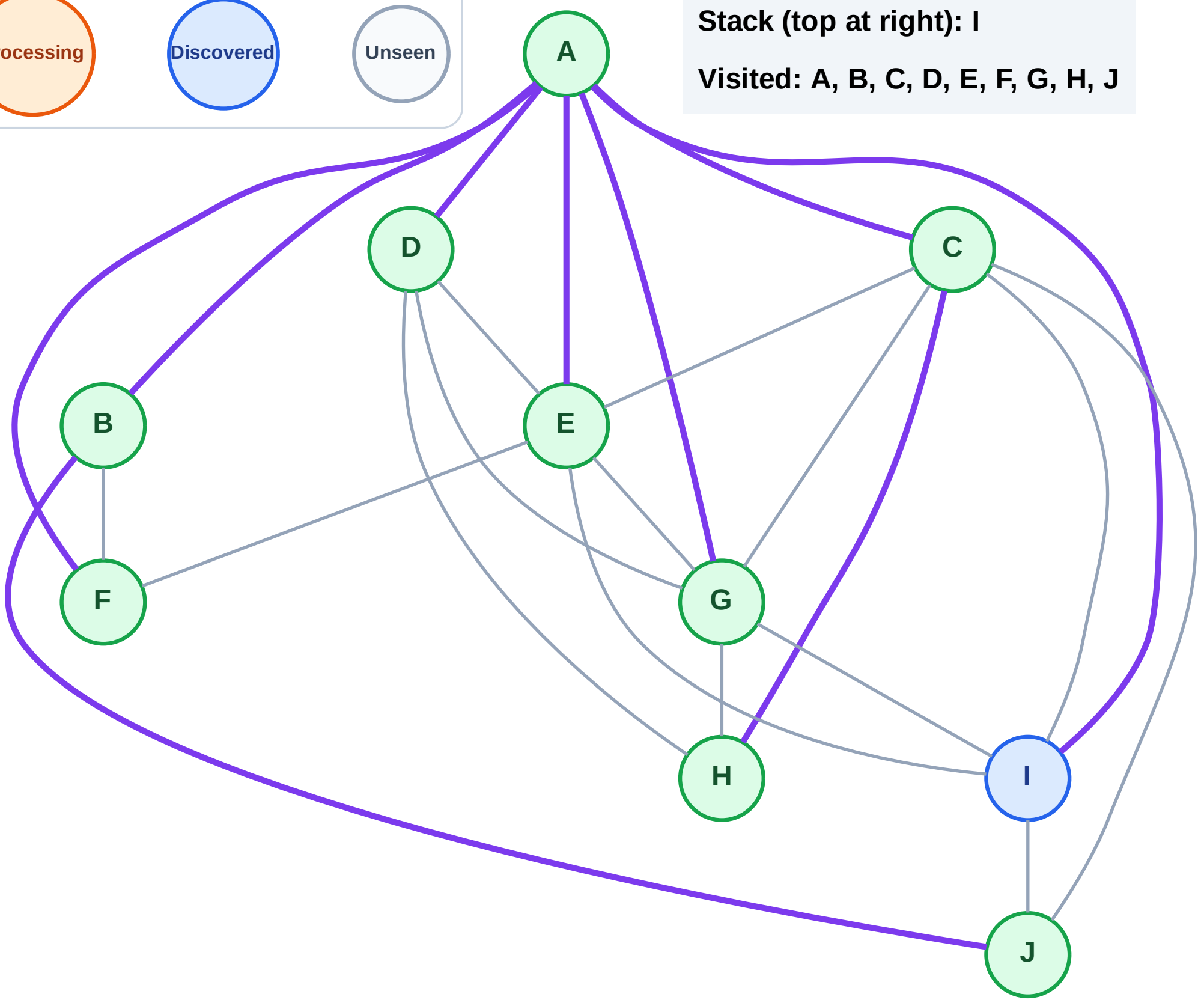
Step 19: No new neighbors from G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

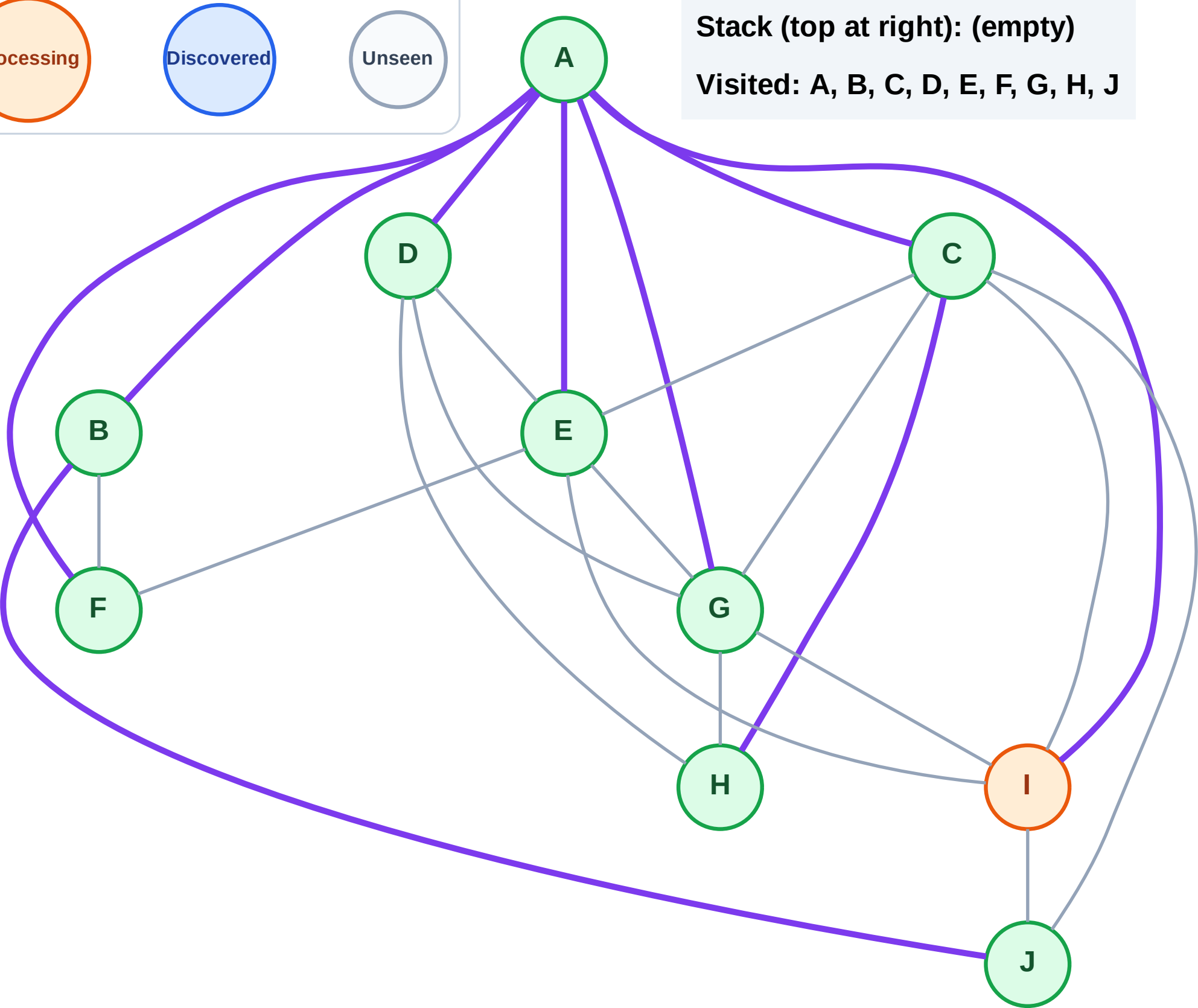
Complete

Processing

Discovered

Unseen

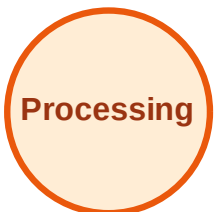
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

